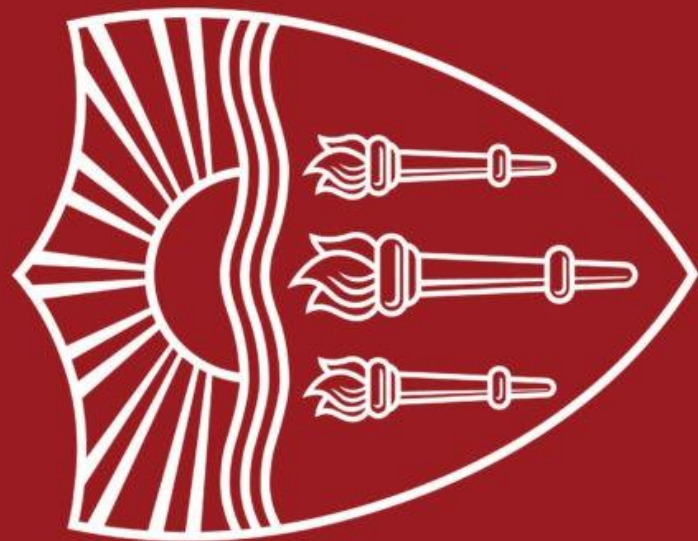


Lecture 1: Introduction and Course Overview

Instructor: Xiang Ren
CSCI 544 Applied NLP
Fall 2026





OpenAI Is Testing an A.I.-Powered Search Engine

Can You Be Emotionally Reliant on an A.I. Voice? OpenAI Says Yes.

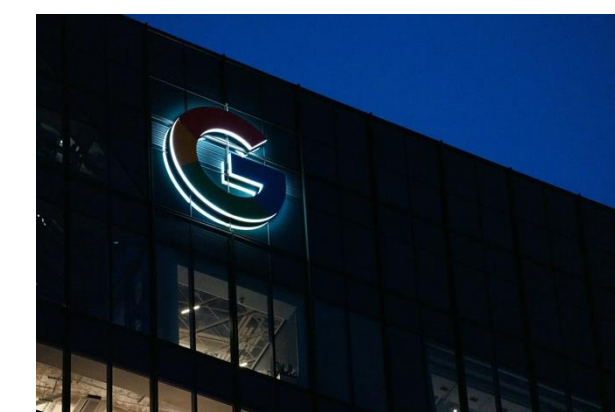
A.I.'s Insatiable Appetite for Energy



Meta, in Its Biggest A.I. Push, Places Smart Assistants Across Its Apps

In Constant Battle With Insurers, Doctors Reach for a Cudgel: A.I.

A.I. Can Write Poetry, but It Struggles With Math



THE SHIFT
Google C.E.O. Sundar Pichai on the A.I. Moment: 'You Will See Us Be Bold'
In an extended interview, Mr. Pichai expressed both optimism and worry about the state of the A.I. race.

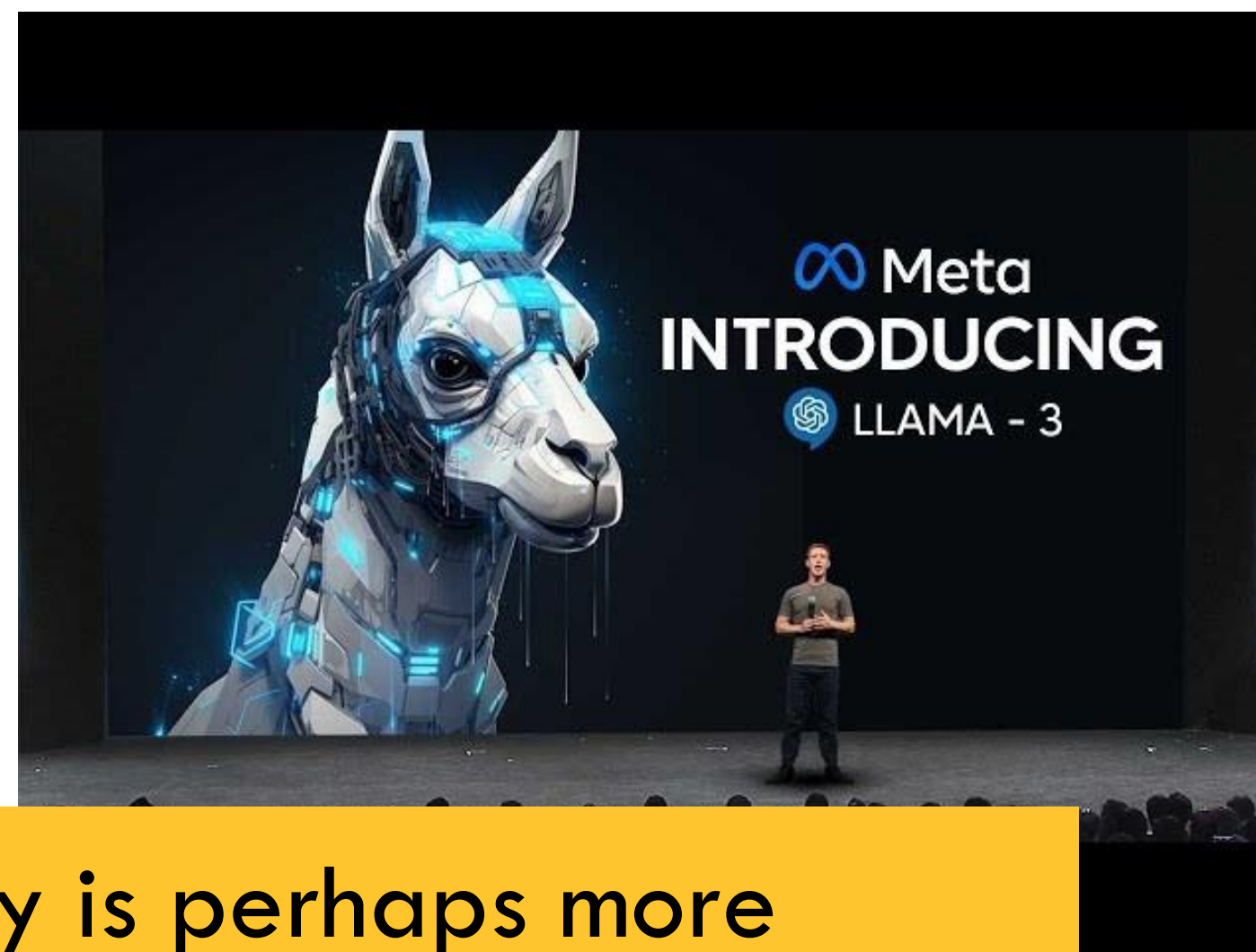
Aided by A.I. Language Models, Google's Robots Are Getting Smart

In Big Election Year, A.I.'s Architects Move Against Its Misuse

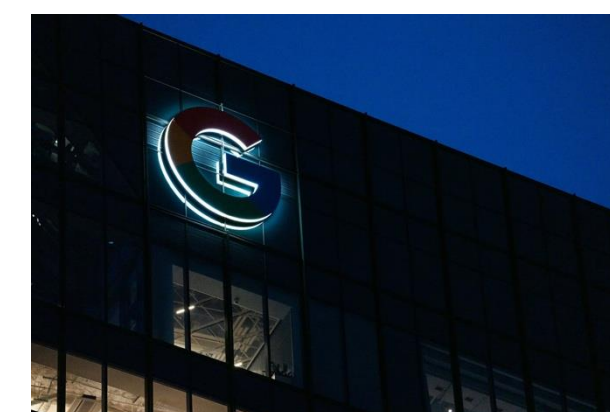
The Chef Is Human. The Reviewer Isn't.



OpenAI Is Testing an A.I.-Powered Search Engine



NLP today is perhaps more relevant than it ever has been!



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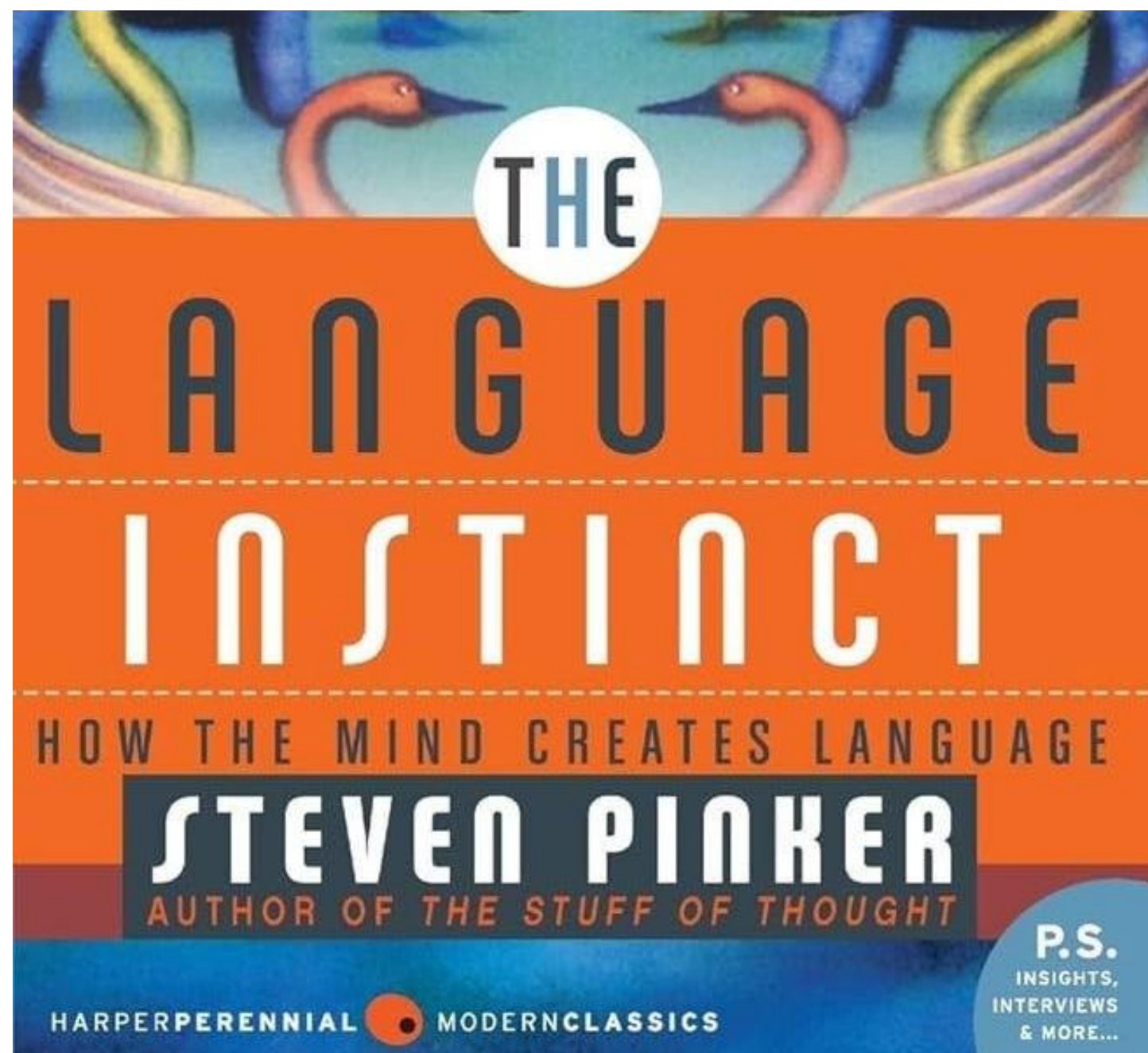
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led by A.I. Language Models, Google's Robots Are Getting Smart

Language is a key modality



“Humans are so innately hardwired for language that they can no more suppress their ability to learn and use language than they can suppress the instinct to pull a hand back from a hot surface.”

– Steven Pinker

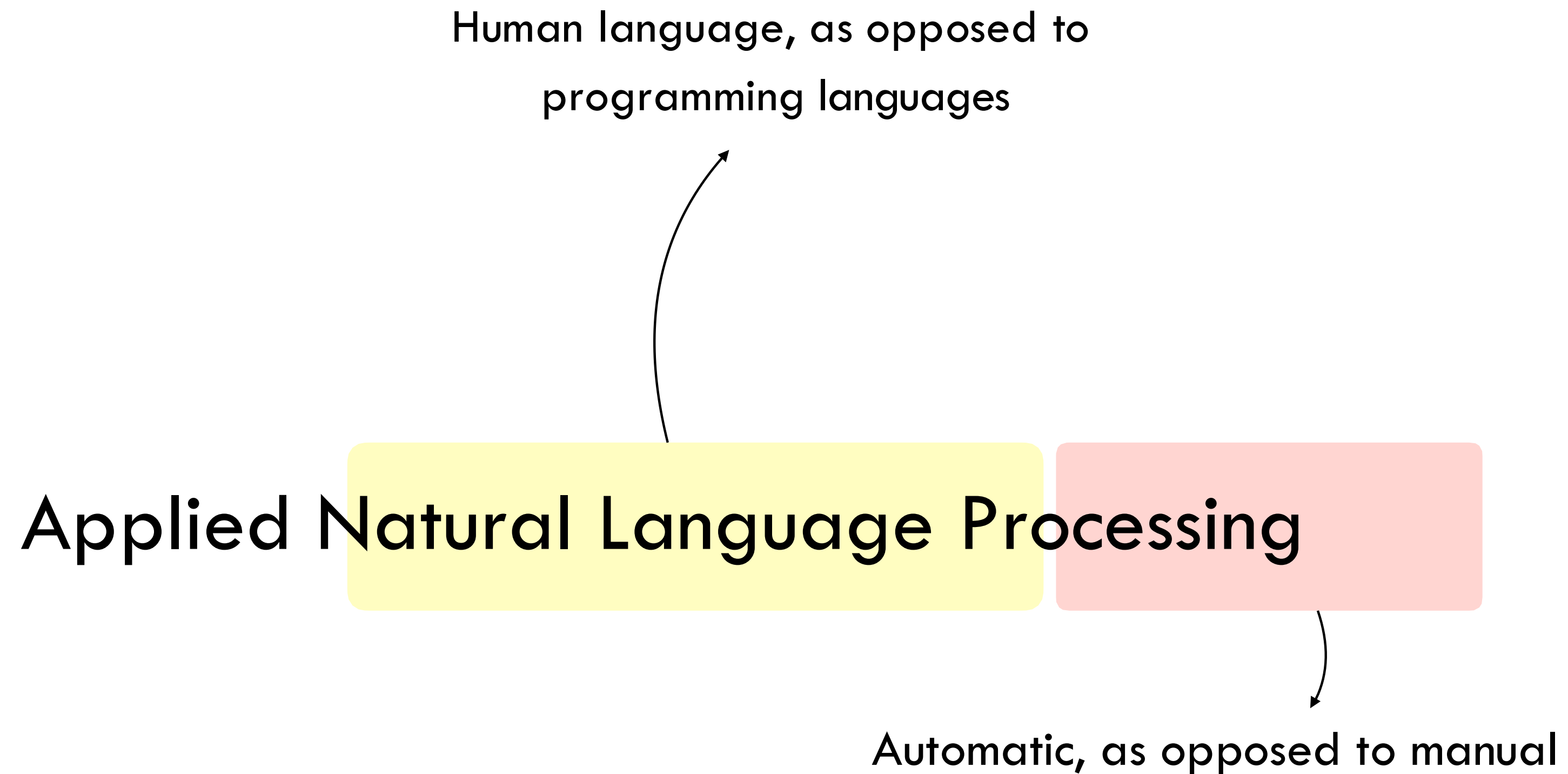
Four
Minute
Books

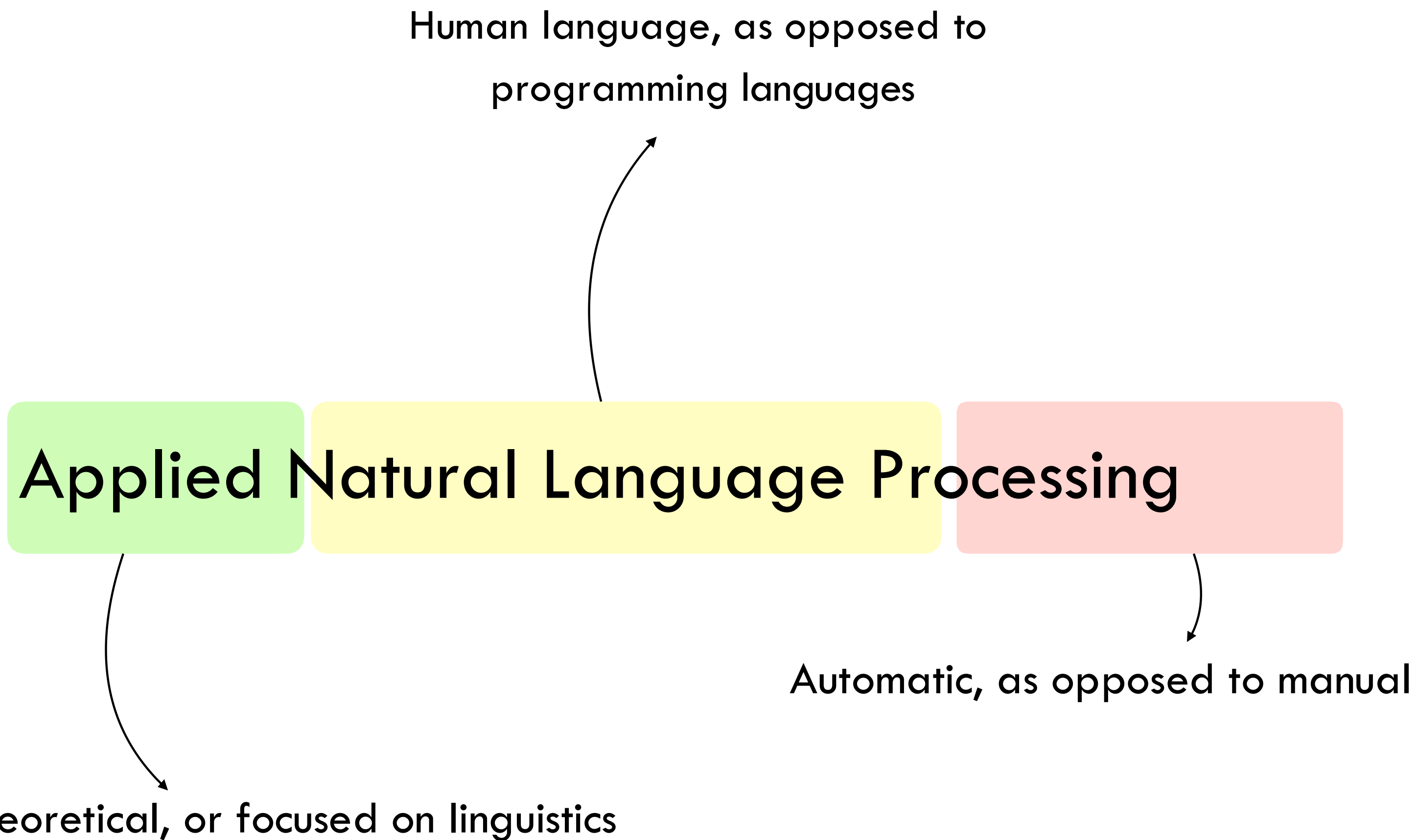
Applied Natural Language Processing

Applied Natural Language Processing



Automatic, as opposed to manual



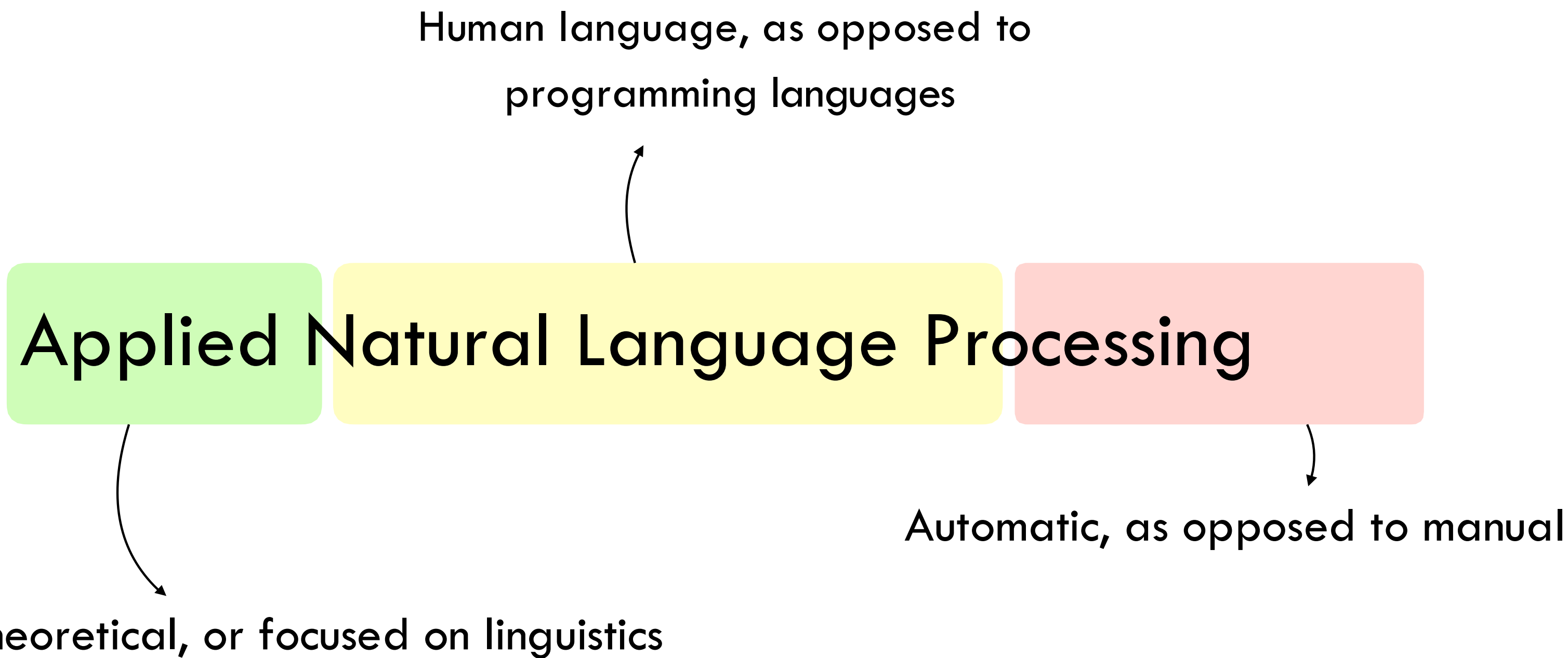


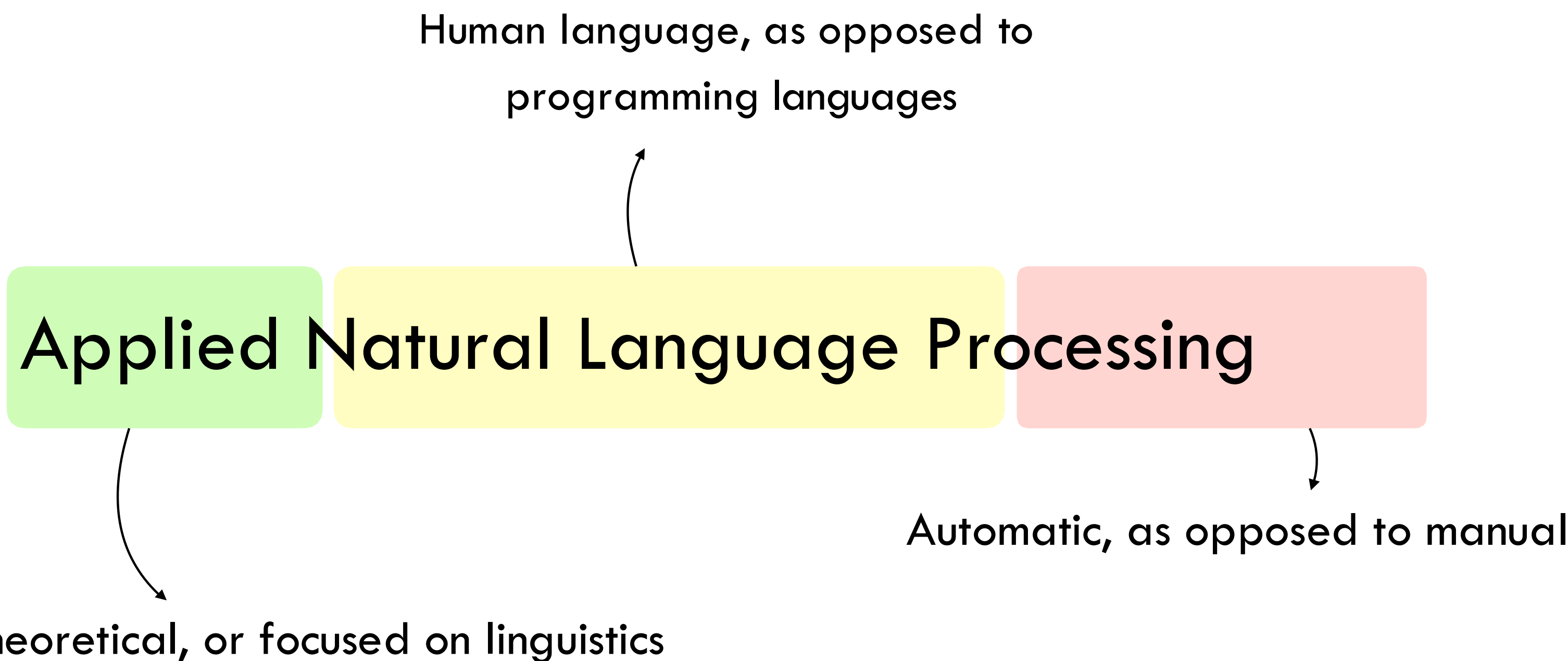
What is

Natural Language

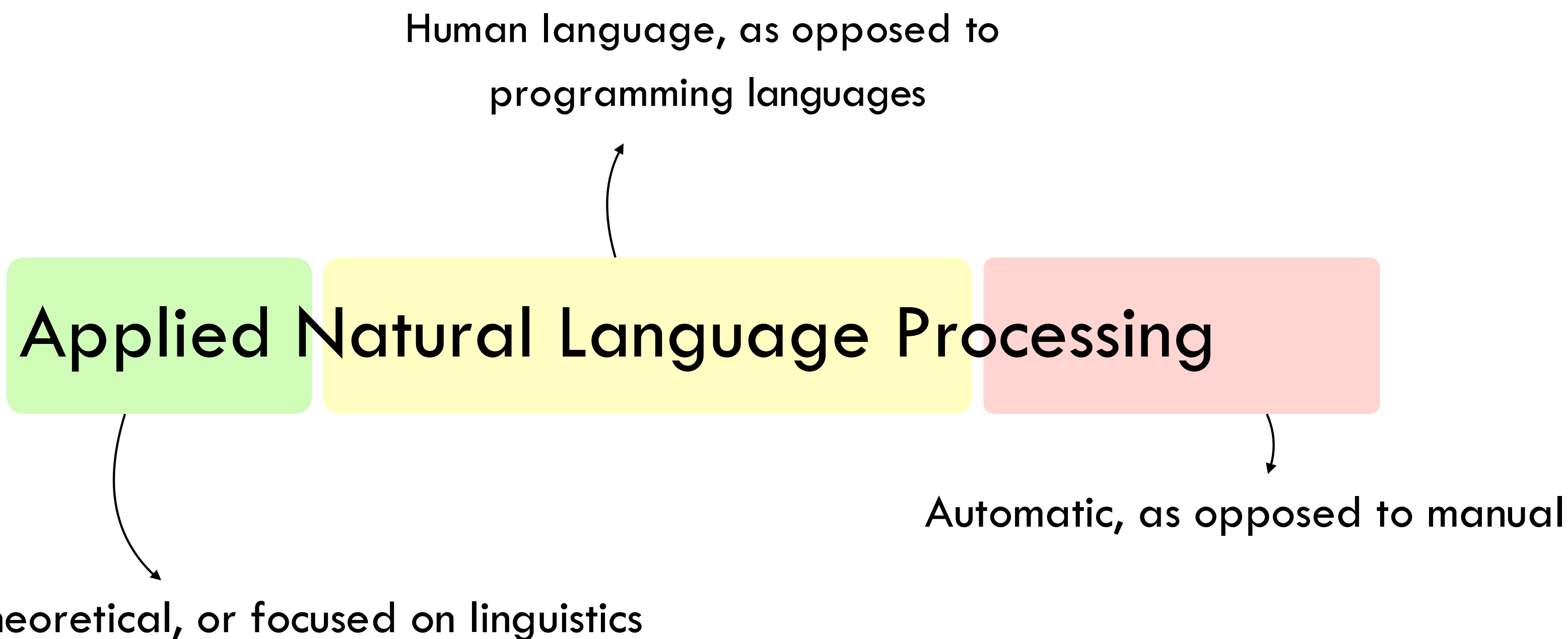
Processing ?

- Field at the intersection of computer science, AI (especially machine learning) and linguistics
- Goal: for computers to process human language, similar to human understanding, towards performing useful tasks
- Challenge: understanding and representing the meaning of language is something even humans struggle with
- Processing: produce outputs (Y) with language or text as input (X)





Applied NLP = Almost entirely Language Models



Applied NLP = Almost entirely Language Models

More generally, language + Z

A General Framework for NLP Systems

- Create a function to map an **input** X into an **output** Y , where X and/or Y involve language.

Input X

Text

Text

Text

Text

Image

Output Y

Continuing Text

Text in Other Language

Label

Linguistic Structure

Text

Task

Language Modeling

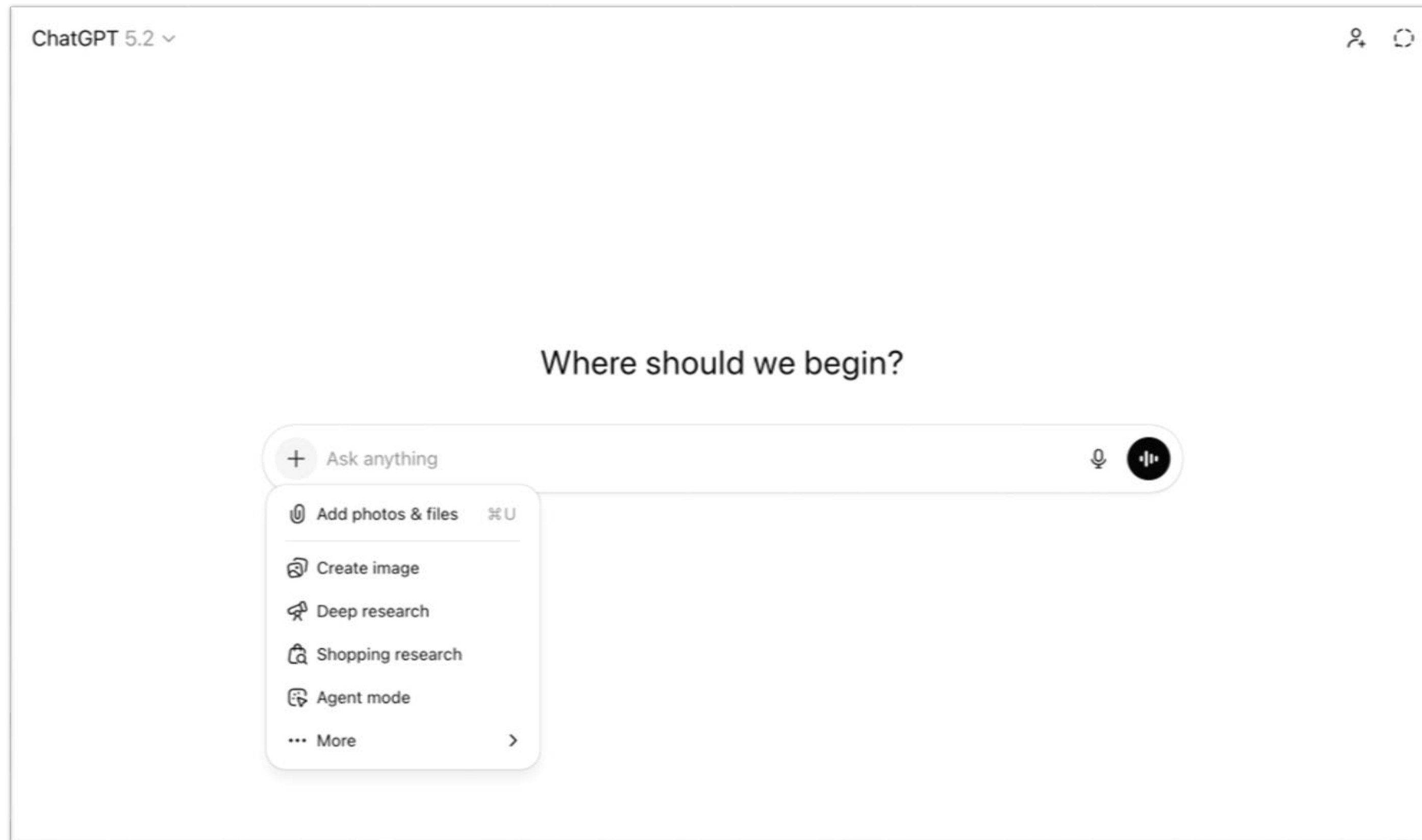
Translation

Text Classification

Language Analysis

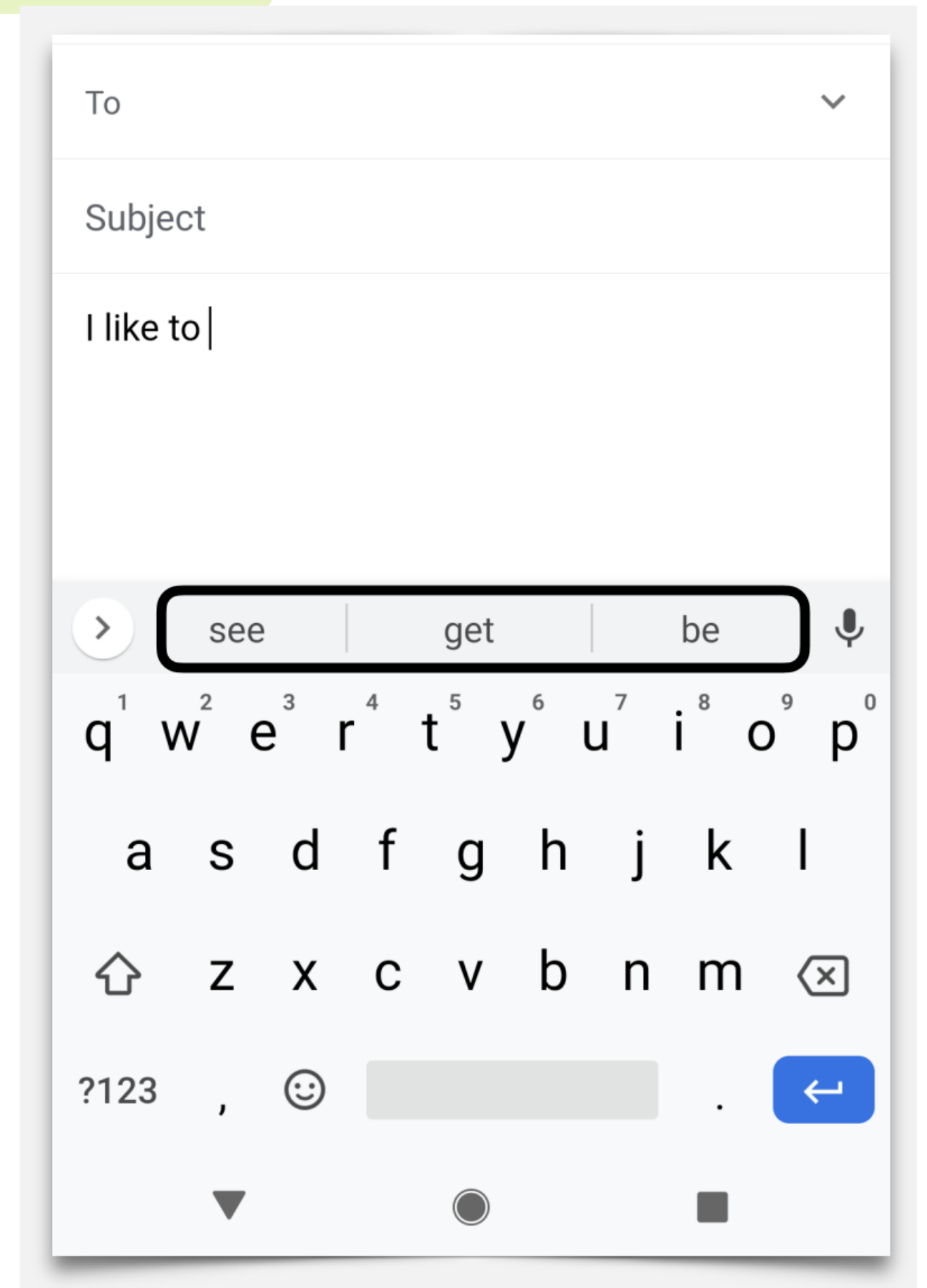
Image Captioning

Today's NLP



Language Models

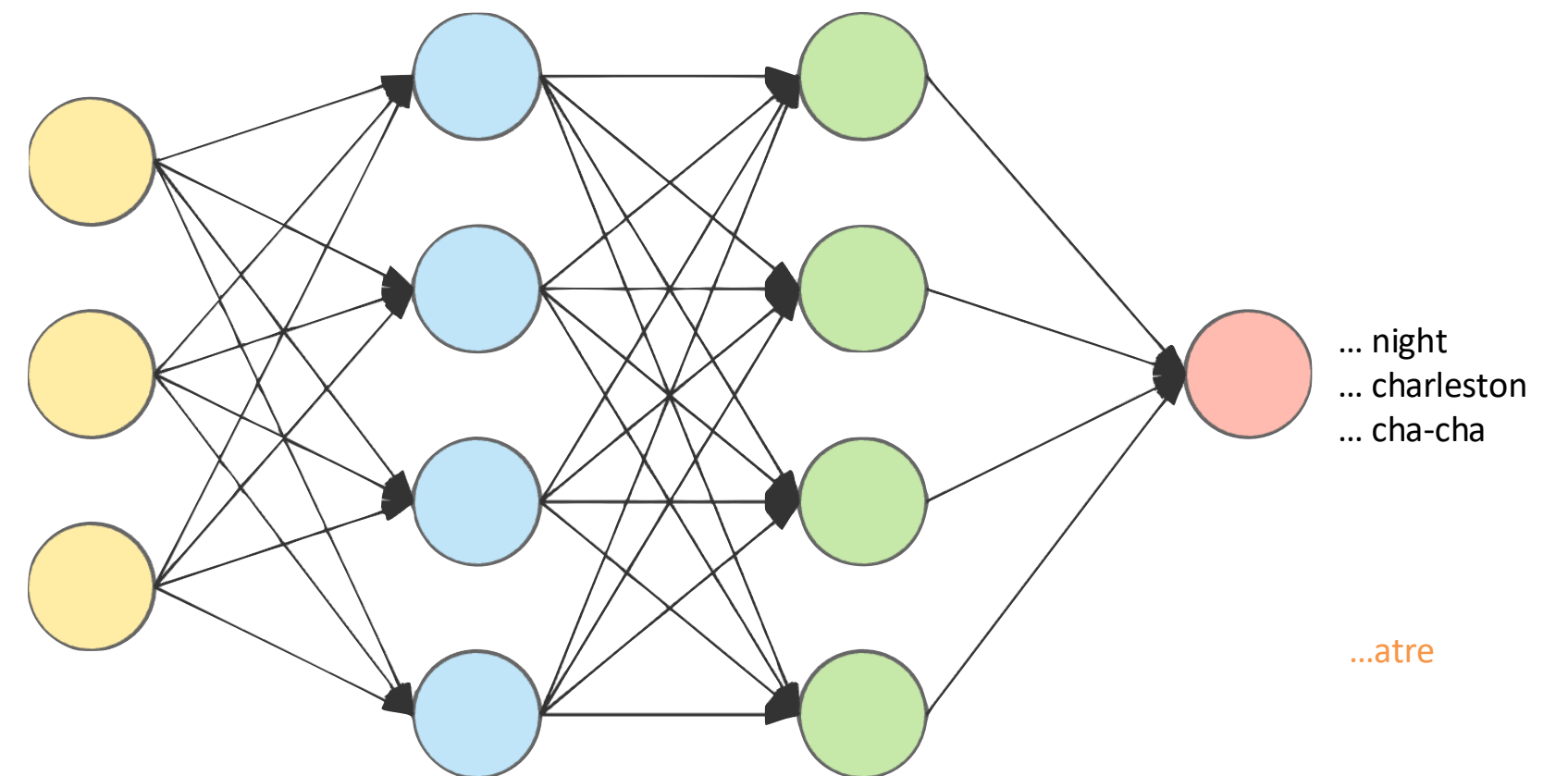
- Task: Given a sequence of words so far (the context), predict what comes next
 - Like autocomplete!
- We never know for sure what comes next, but we can still make good guesses!
- Question: what is X and what is Y here?
 - Input X = Language and Output Y = Language



Language Model as Autocomplete

Task: Given a sequence of words so far
(the context), predict what comes next
We never know for sure what comes next,
but we can still make good guesses!

Dance the...



What words can follow this?



What is common to these words?

Language Model as Autocomplete

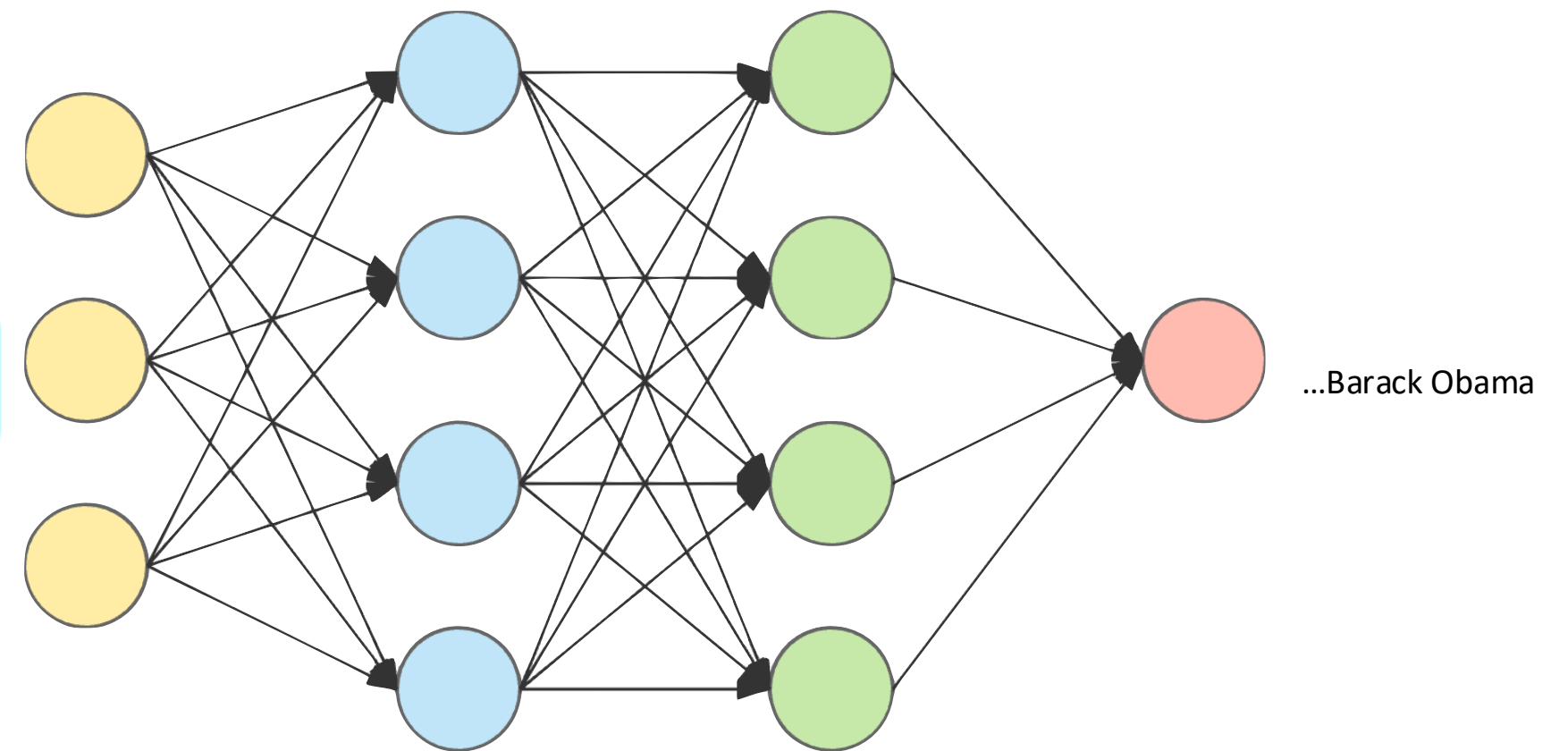
Task: Given a sequence of words so far (the context), predict what comes next

We never know for sure what comes next, but we can still make good guesses!

**The 44th President of
United States was ...**



What words can follow this?



I want to _____

◀

!

|

Language Models

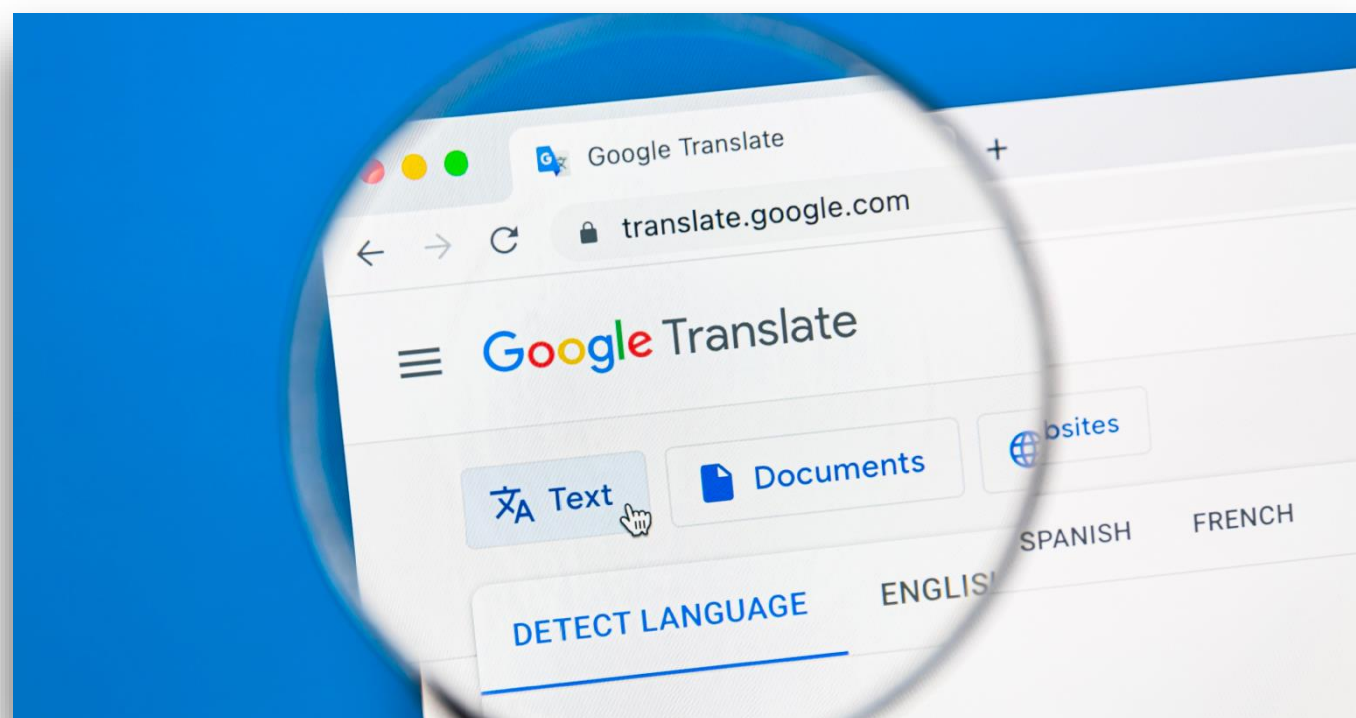
- But today, variants of language models are used
 - As chat models
 - As supercharged auto completes
 - As instruction-following assistants
 - And many more...

Apple's Siri

- Understands the user
- Remembers what the user said earlier
- Can understand which alarm she is referring to



Google Translate



- Detects language automatically
- Can reorder spans in text on the fly

中共中央政治局7月30日召开会议，会议分析研究当前经济形势，部署下半年经济工作。

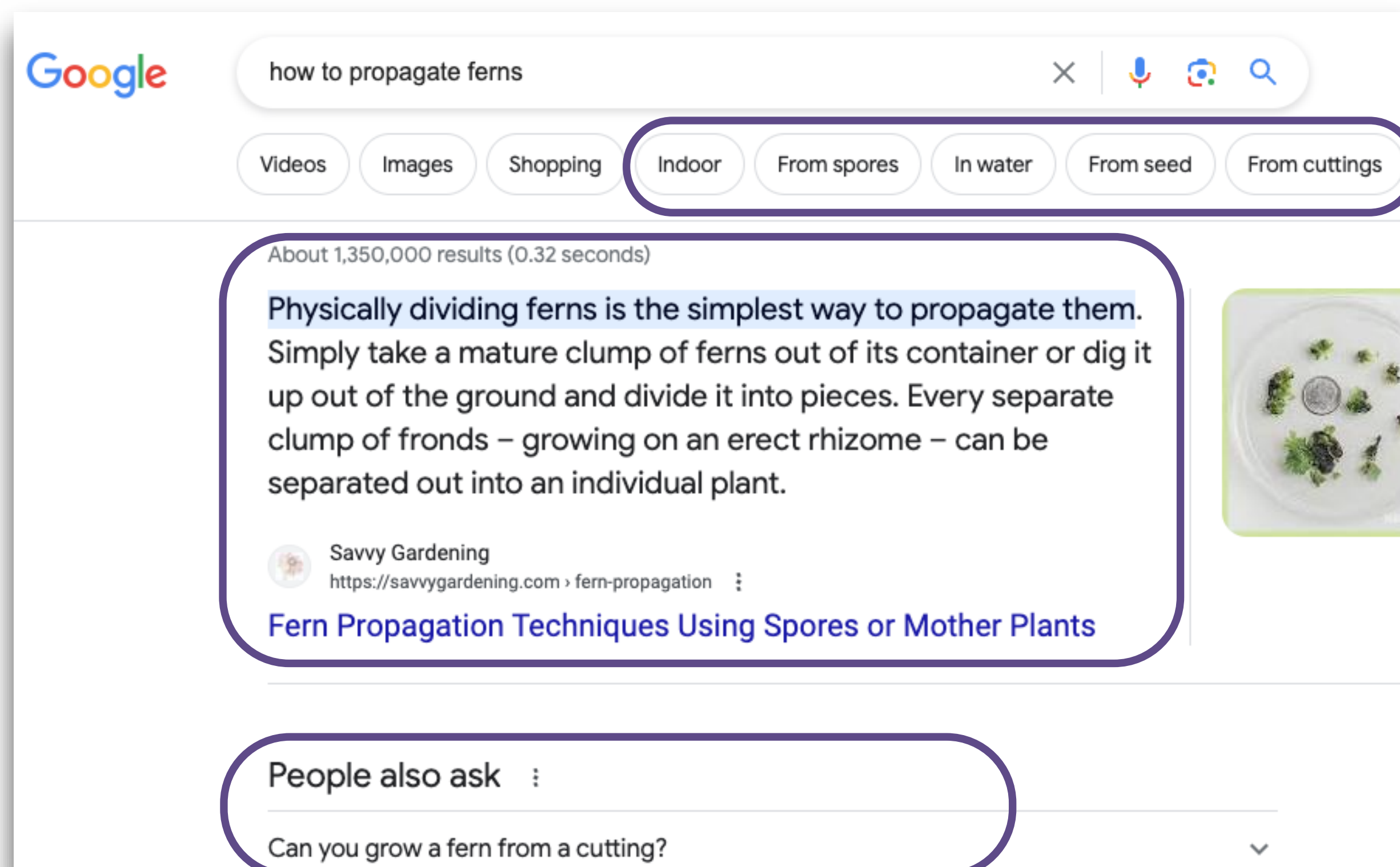
People's Daily, August 10, 2020

Translate

The Political Bureau of the CPC Central Committee held a meeting on July 30 to analyze and study the current economic situation and plan economic work in the second half of the year.

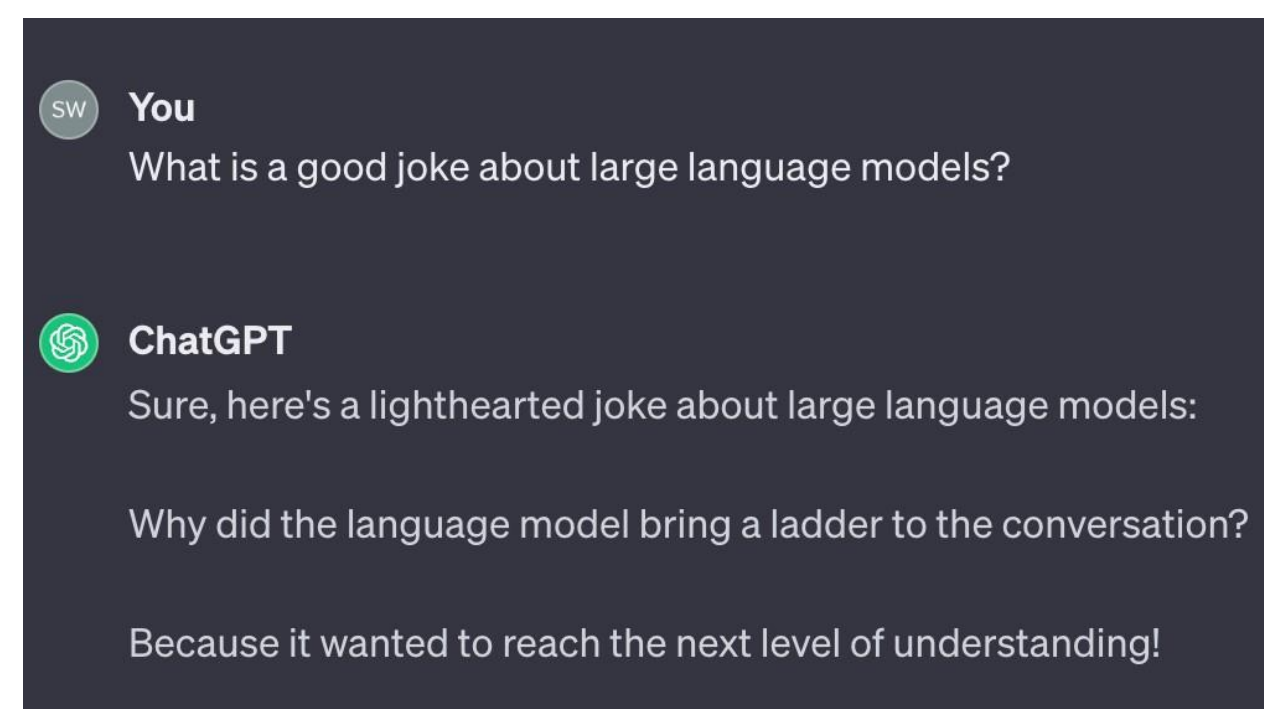
Google Search

- Understands that a fern can be indoor, can be propagated either from seed or from cuttings
- Can find the exact passage in a webpage that answers the questions
- Can find related (in meaning) questions



Language Models

- But today, variants of language models are used
 - As chat models
 - As supercharged auto completes
 - As instruction-following assistants
 - And many more...
- It's hard to know exactly how these models might respond, making these models stochastic / probabilistic
 - Contrast this with rule-based systems which respond exactly the same way each time (deterministic systems)



Lecture Outline

- What is Applied NLP today?
 - Language Models!
- Why should we care about language models?
- Class Syllabus
- Class Logistics

Instructor, Website and Students



Instructor: Xiang Ren

xiangren@usc.edu

Office hours: Friday 9:00-10:00 AM, GCS SB5 (under Level 2)



TA: Huijuan Wang

huijuanw@usc.edu



TA: Xu Wang

xwang032@usc.edu



TA: Shicheng Wen

wenshich@usc.edu



TA: Hirona Arai

hjarai@usc.edu

Why study Applied NLP?

~~Why study Applied NLP?~~

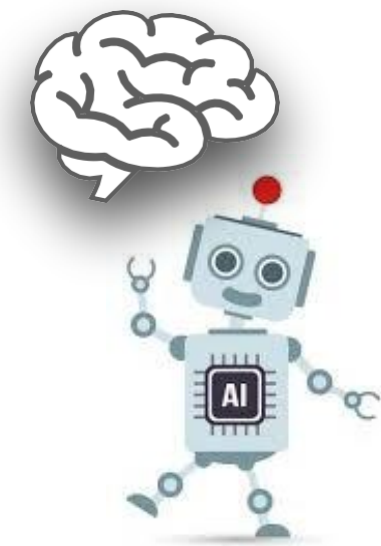
Why study Language Models?

Related Question: Why study AI?

The future is
Artificial Intelligence

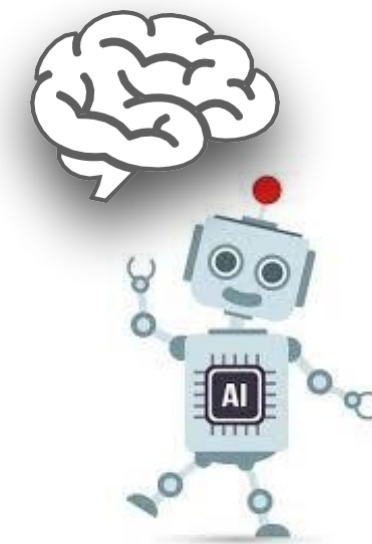
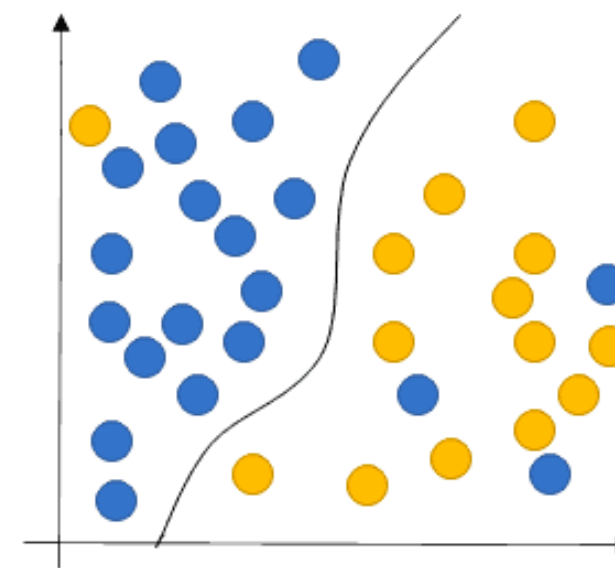


Artificial Intelligence



Artificial Intelligence

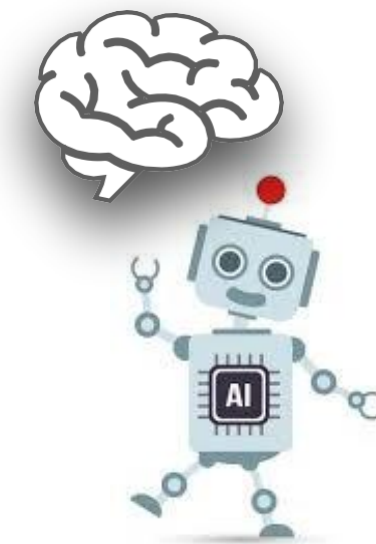
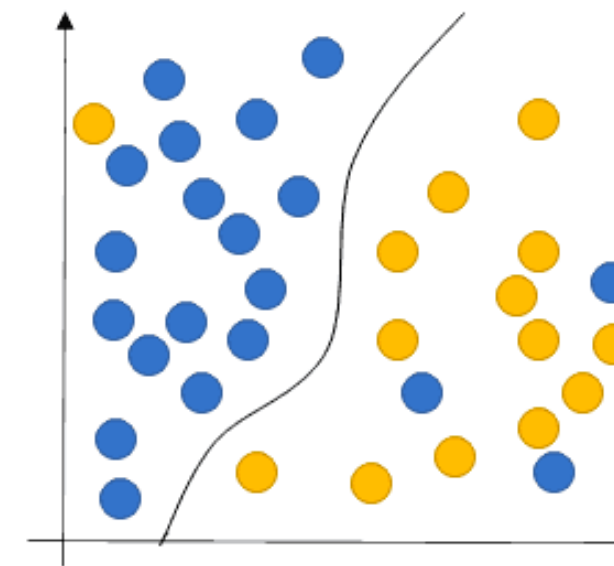
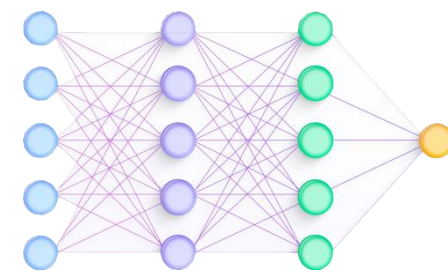
Machine Learning and Natural Language Processing



Artificial Intelligence

Machine Learning and Natural Language Processing

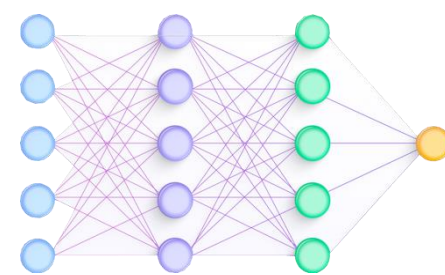
Deep Learning



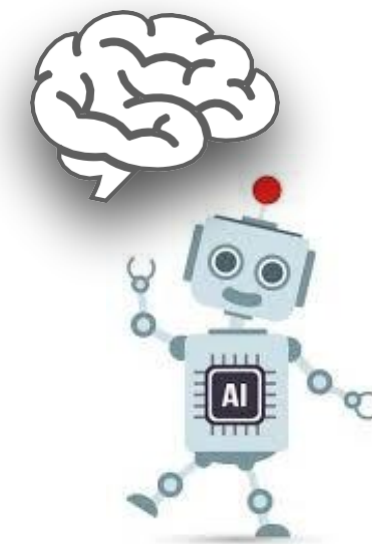
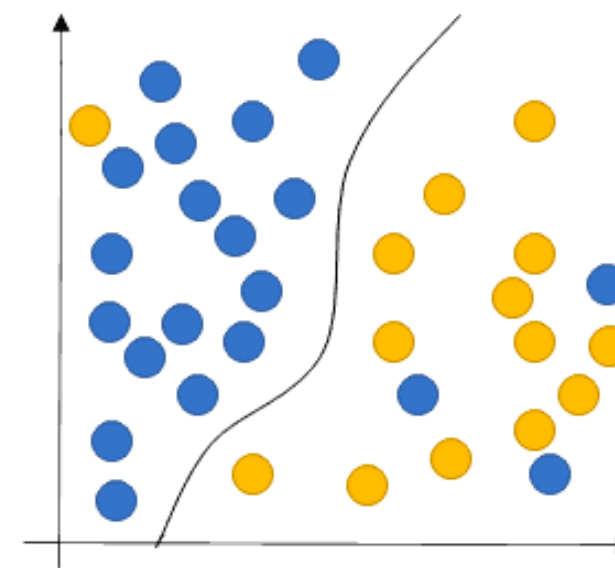
Artificial Intelligence

Machine Learning and Natural Language Processing

Deep Learning



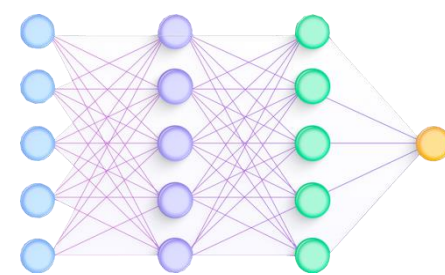
Generative AI



Artificial Intelligence

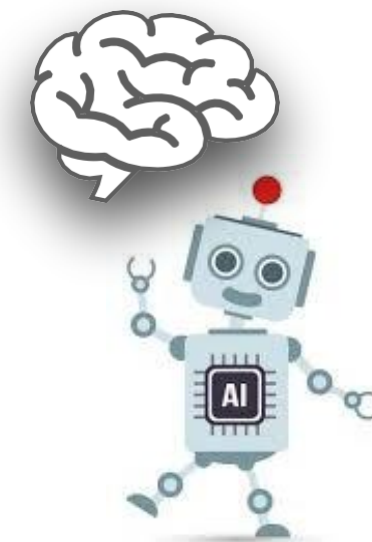
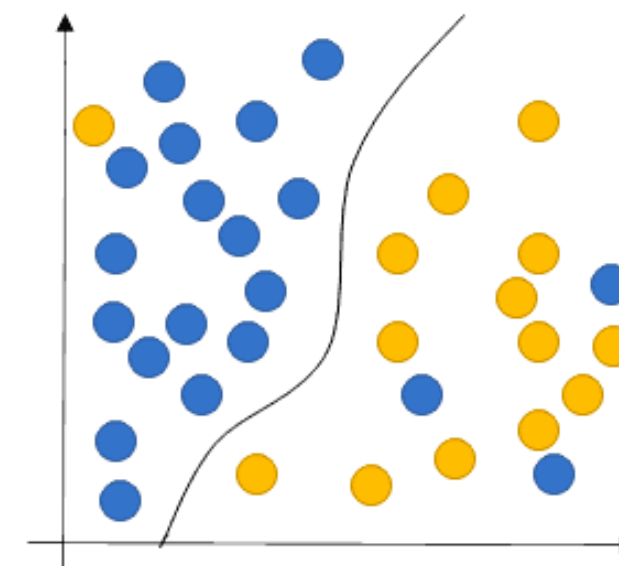
Machine Learning and Natural Language Processing

Deep Learning



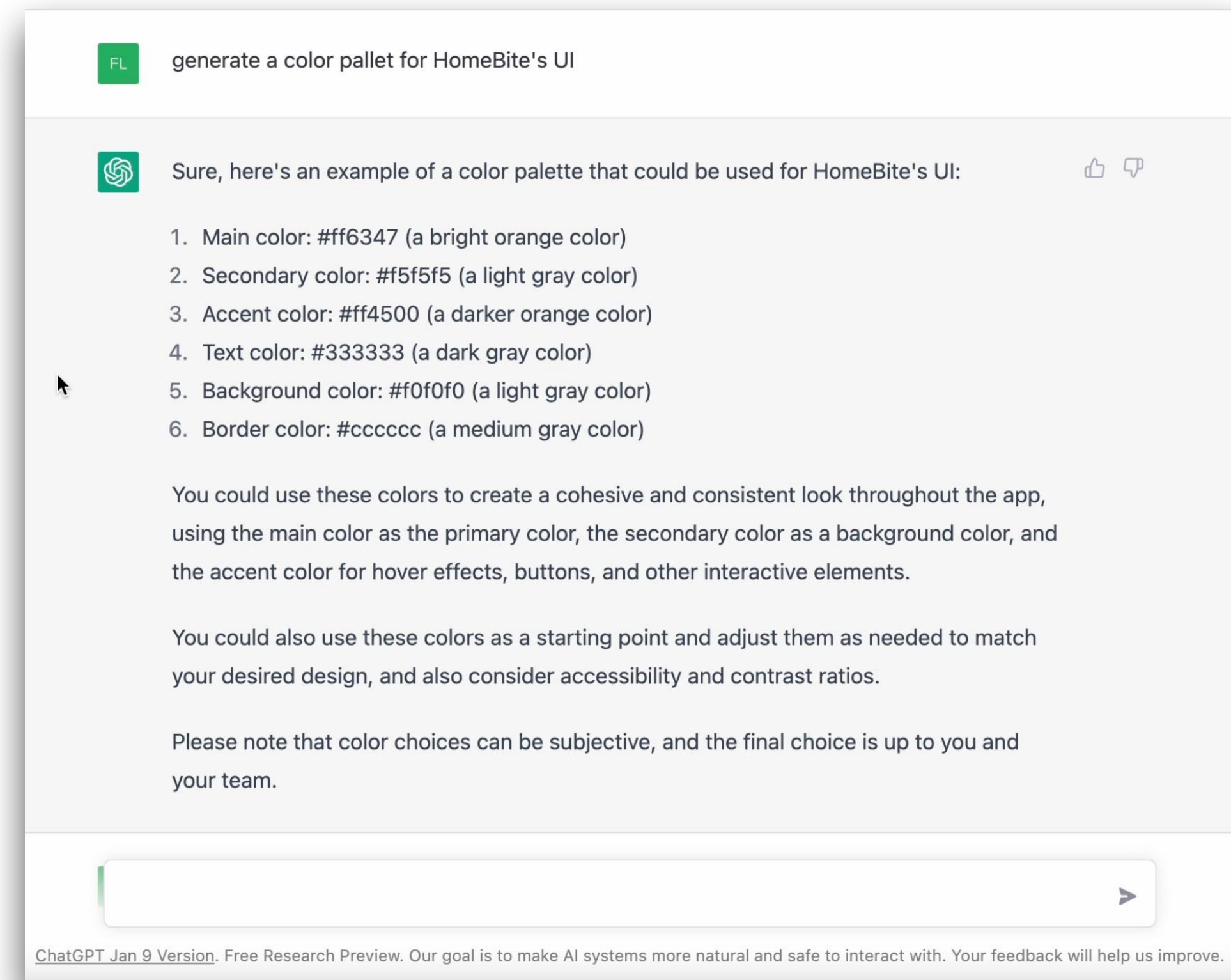
Generative AI

Language
Models



Why should we care about language models?

- Foundational models in NLP today
 - Internal representations of models is learned exclusively from sequence/textual data
 - Direct applications of LLMs are growing rapidly
- Classical importance
- Ubiquitous, so everyone has an opinion!
- Immense societal implications



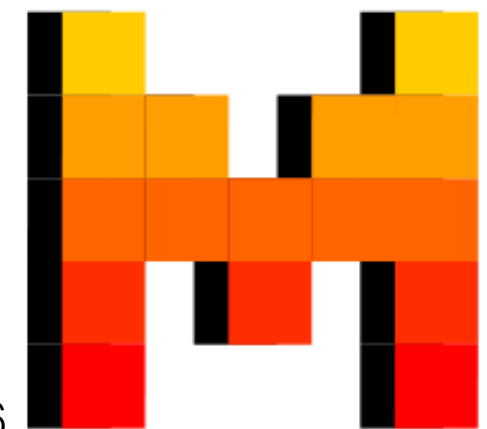
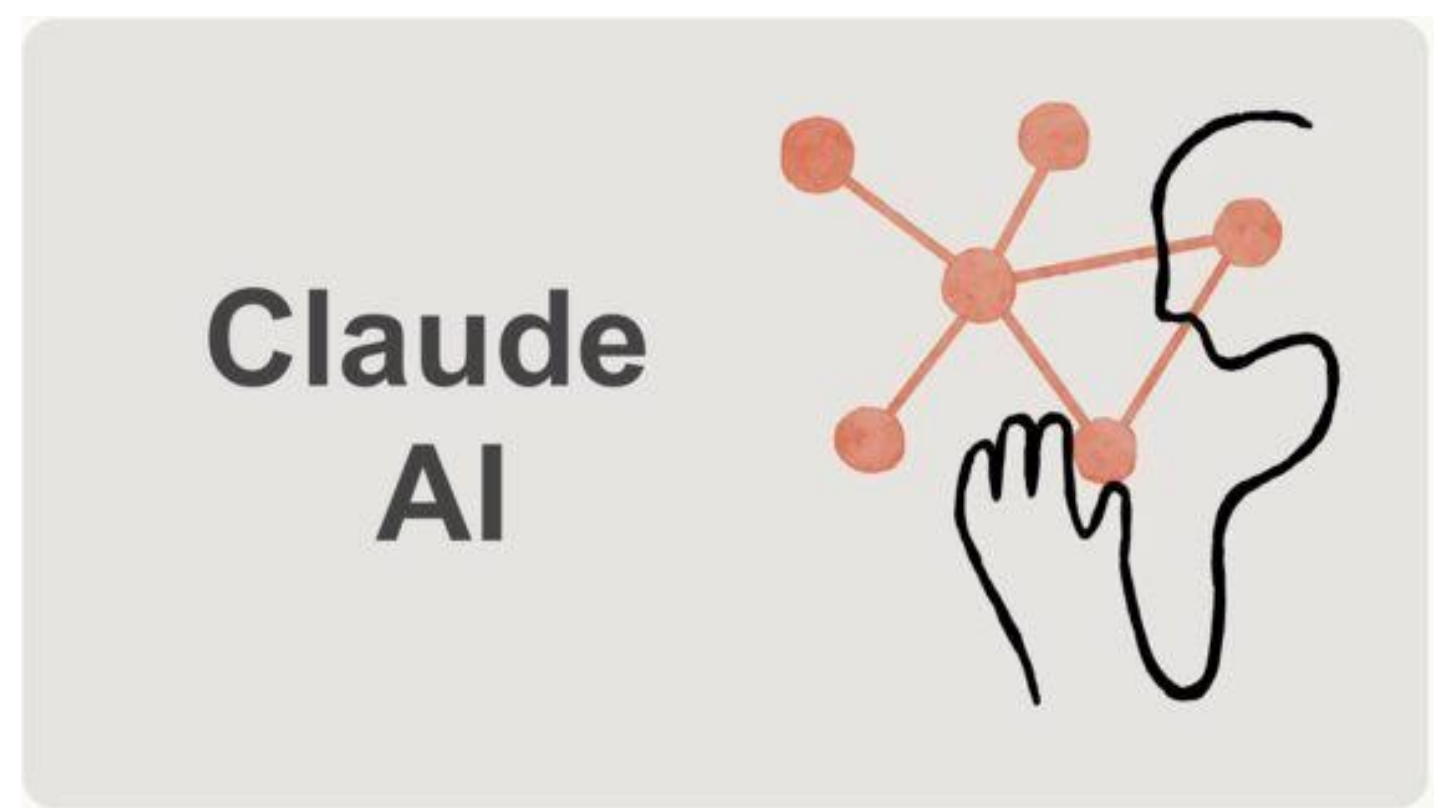




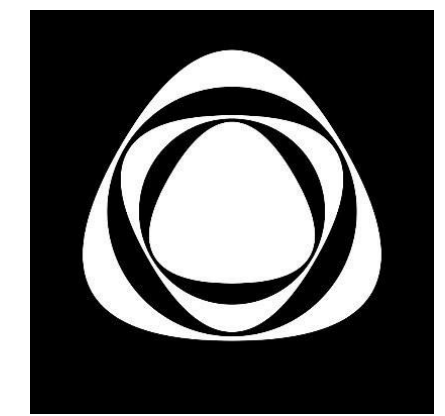
GPT - 4



LLM360



MISTRAL
AI_





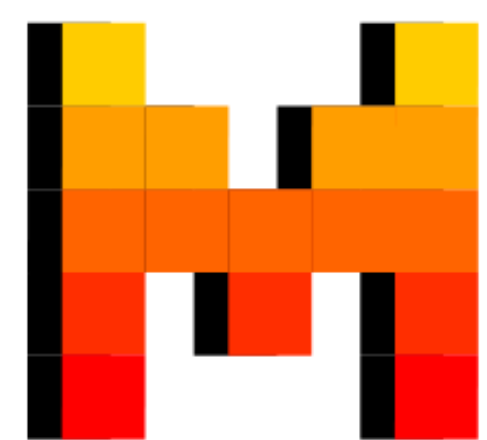
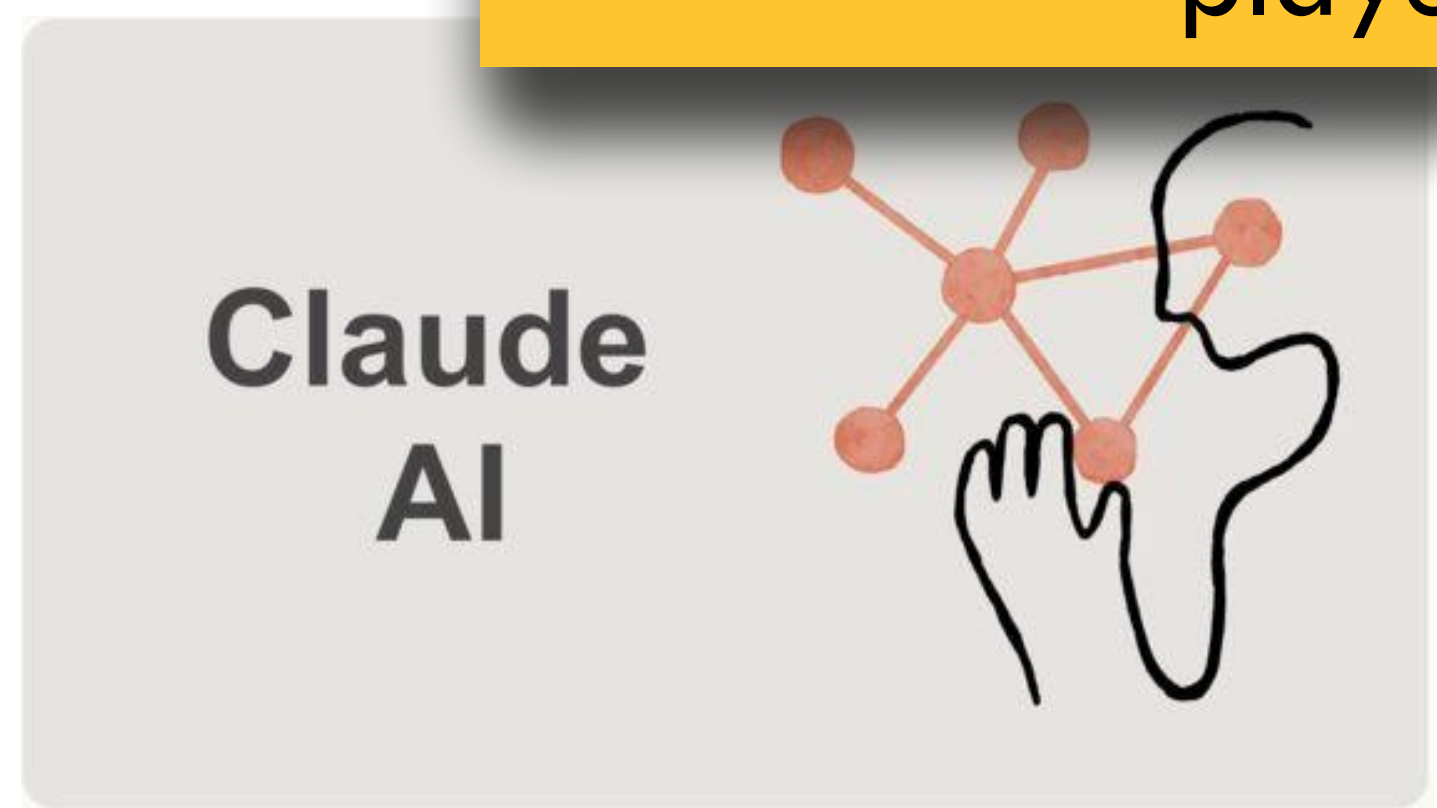
GPT-4



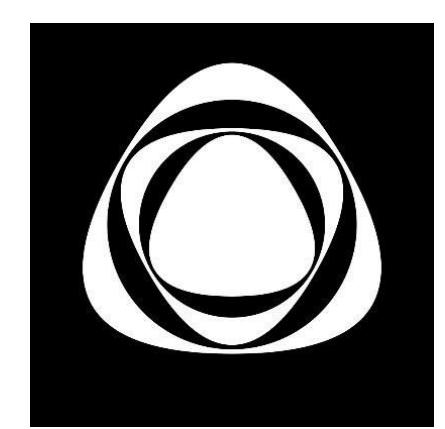
LLM360



Language models are getting larger, only a few key players can develop them



MISTRAL
AI_





**Virtual
Assistants**



Translation



**Content
Creation**



**Code
Generation**



Transcription



**Sentiment
Analysis**

Large Language Models Are Everywhere



**Virtual
Assistants**



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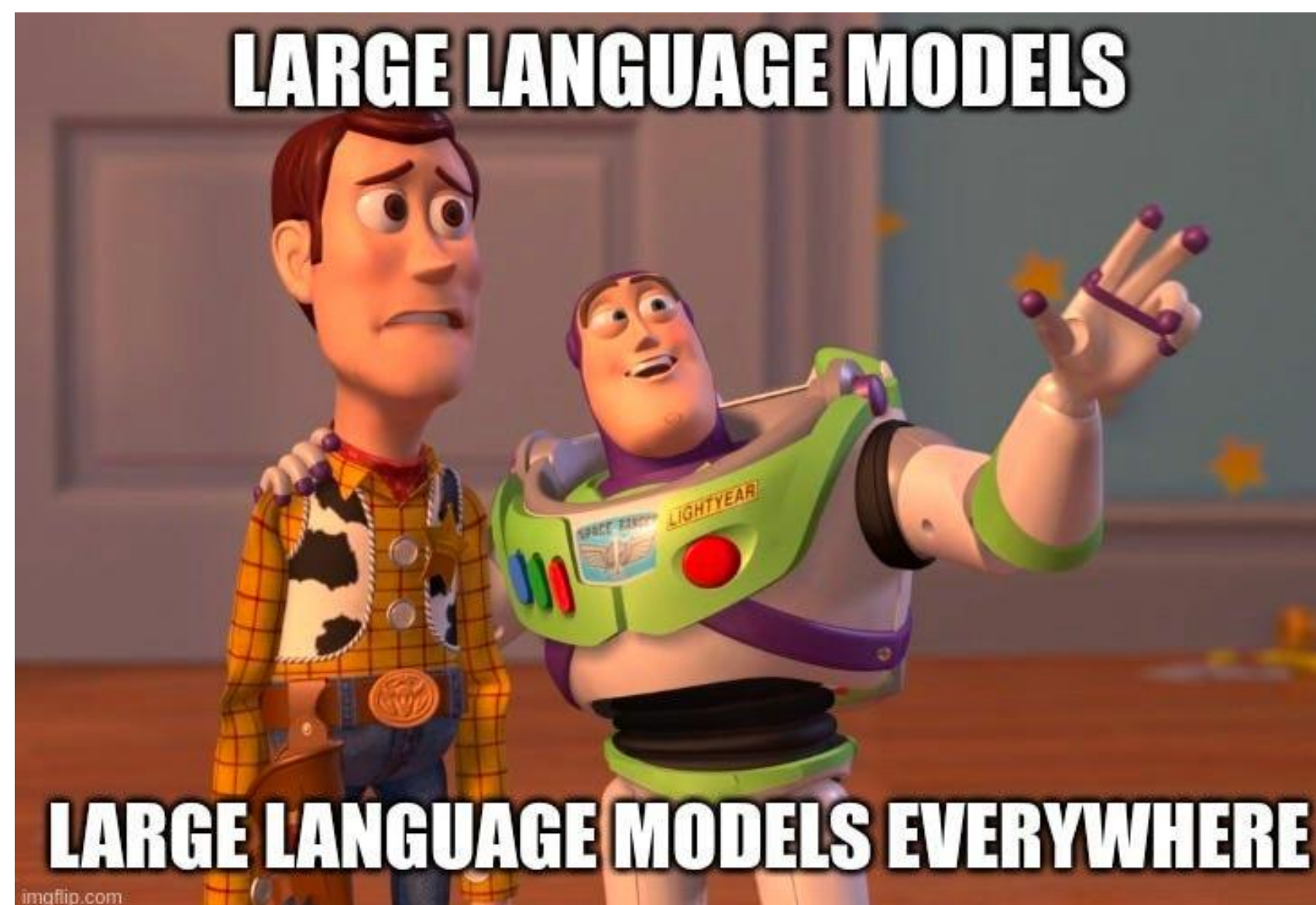
Code
Generation



Transcription

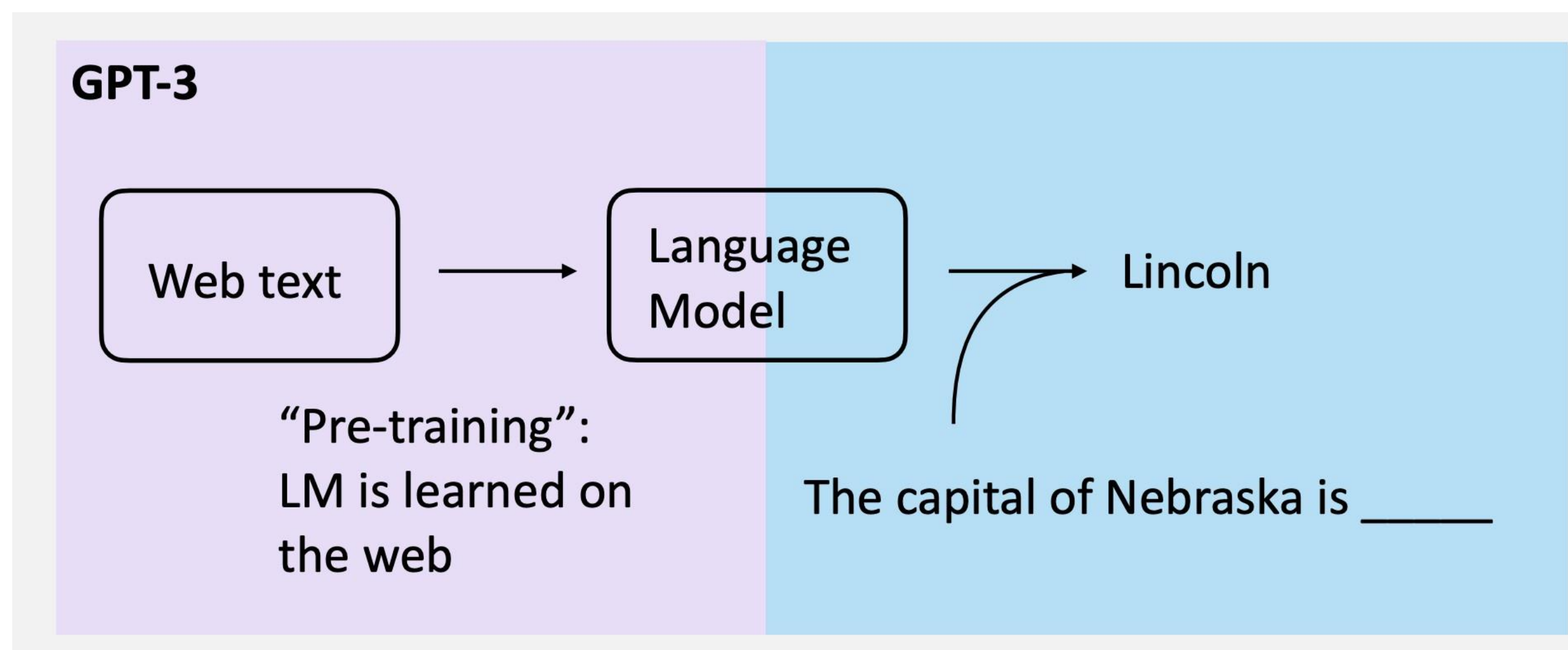


Sentiment
Analysis



Large Language Models (LLMs)

- Models are trained tons (terabytes) of data — a curated scrape of the web, and even generated data
- Have seen information about common and even not-so-common entities
- A big enough model can answer questions even without being trained to do so. What else can we get these models to do?



LLMs are being used to...

LLMs are being used to...

Code computer programs!

Summarize articles, podcasts or presentations

Draft emails

Script social media posts

Create a title for an article

Play games

Assist with job searches,
including writing resumes
and cover letters

Ask trivia questions

Compose music!!!

Discover keywords for search
engine optimization.

Formulate product
descriptions

Describe complex topics more simply

Solve math problems

Create articles, blog posts
and quizzes for websites

Reword existing content for a different medium, such
as a presentation transcript for a blog post

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Assist with job searches
including
and c

Ask trivia questions

Generate images

**Extremely powerful: can in many cases replace
laborious manual efforts**

Describe complex topics more simply

engine optimization.

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Solve math problems

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Reword existing content for a different medium, such
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GPT-4 Passes the Bar Exam!

GPT Takes the Bar Exam

December 29, 2022

Michael Bommarito II^{1,2,3}, Daniel Martin Katz^{1,2,3,*}

1 Illinois Tech - Chicago Kent College of Law (Chicago, IL USA)

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* Corresponding Author: dkatz3@kentlaw.iit.edu

Abstract

Nearly all jurisdictions in the United States require a professional license exam, commonly referred to as “the Bar Exam,” as a precondition for law practice. To even sit for the exam, most jurisdictions require that an applicant completes at least seven years of post-secondary education, including three years at an accredited law school. In addition, most test-takers also undergo weeks to months of further, exam-specific preparation. Despite this significant investment of time and capital, approximately one in five test-takers still score under the rate required to pass the exam on their first try. In the face of a complex task that requires such depth of knowledge, what, then, should we expect of the state of the art in “AI?” In this research, we document our experimental evaluation of the performance of OpenAI’s TEXT-DAVINCI-003 model, often-referred to as GPT-3.5, on the multistate multiple choice (MBE) section of the exam. While we find no benefit in fine-tuning over GPT-3.5’s zero-shot performance at the scale of our training data, we do find that hyperparameter optimization and prompt engineering positively impacted GPT-3.5’s zero-shot performance. For best prompt and parameters, GPT-3.5 achieves a headline correct rate of 50.3% on a complete NCBE MBE practice exam, significantly in excess of the 25% baseline guessing rate, and performs at a passing rate for both Evidence and Torts. GPT-3.5’s ranking of responses is also highly-correlated with correctness; its top two and top three choices are correct 71% and 88% of the time, respectively, indicating very strong non-entanglement performance. While our ability to interpret these results is limited by nascent scientific understanding of LLMs and the proprietary nature of GPT, we believe that these results strongly suggest that an LLM will pass the MBE component of the Bar Exam in the near future.

	GPT	GPT Top 2	GPT Top 3	NCBE
Evidence	63%	84%	98%	65%
Torts	62%	72%	93%	71%
Civil Procedure	52%	63%	79%	59%
Constitutional Law	49%	67%	87%	72%
Real Property	45%	72%	85%	65%
Contracts	45%	77%	86%	70%
Criminal Law & Procedure	35%	62%	86%	71%

AVERAGE 50% 71% 88% 68%

Table 2. Summary of performance by question category for GPT-3.5 and NCBE-Reported Students

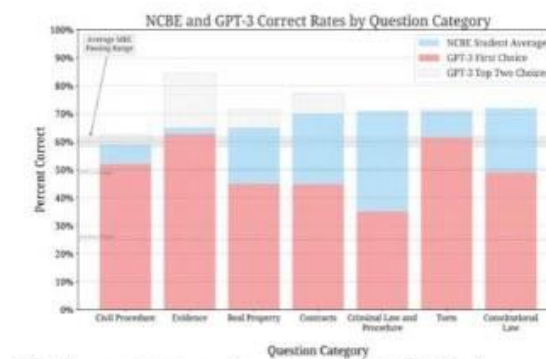


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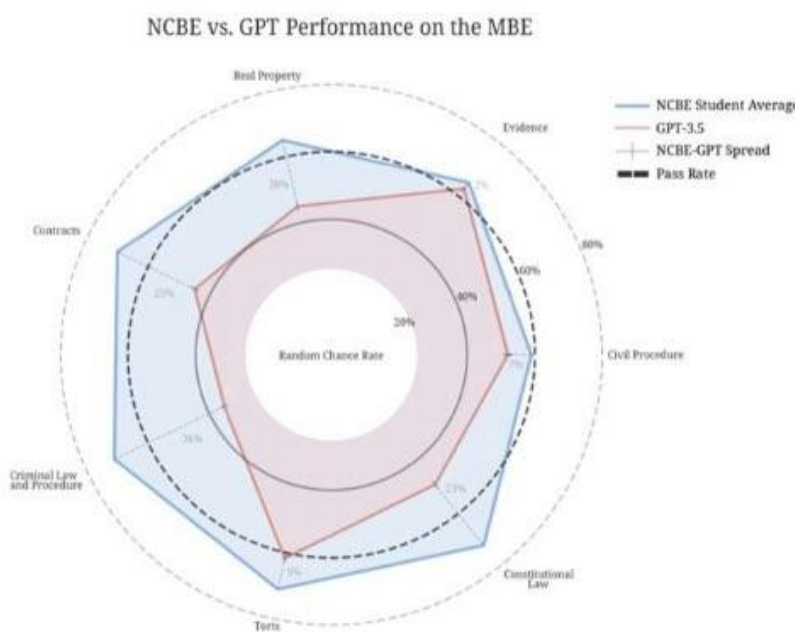
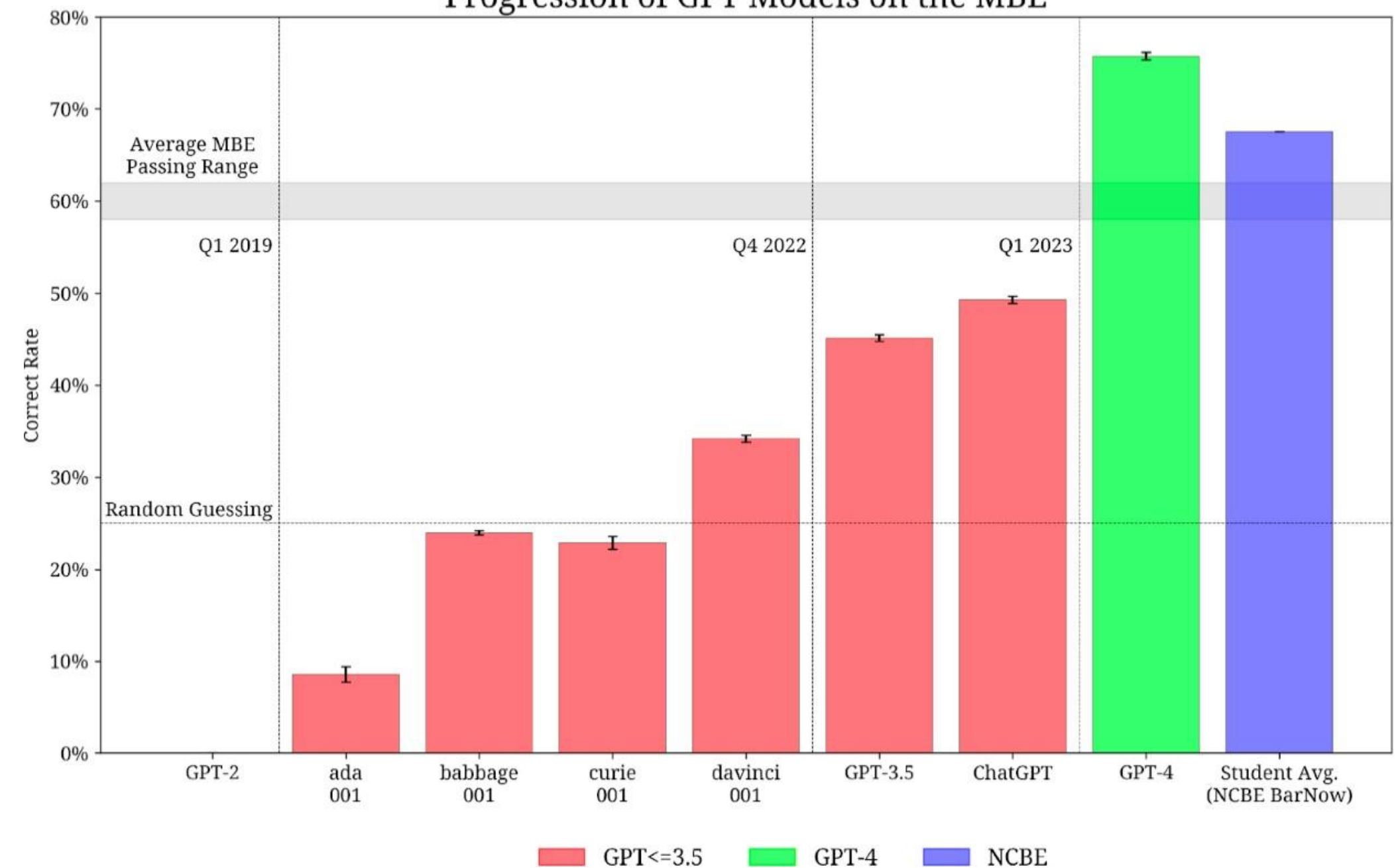


Fig 2. Accuracy by Question Category for GPT and Average Test-Takers

Progression of GPT Models on the MBE



GPT≤3.5 GPT-4 NCBE

Why does this work?

GPT Takes the Bar Exam

Michael Bommarito II^{1,2,3}, Daniel Martin Katz^{1,2,3,*}

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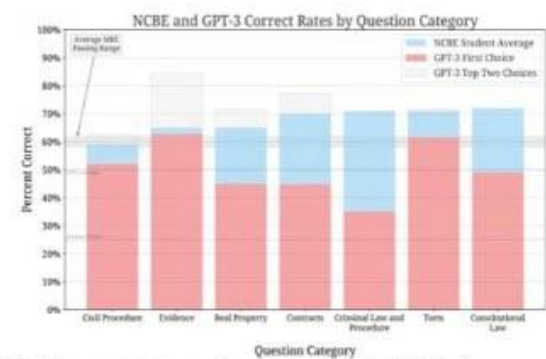


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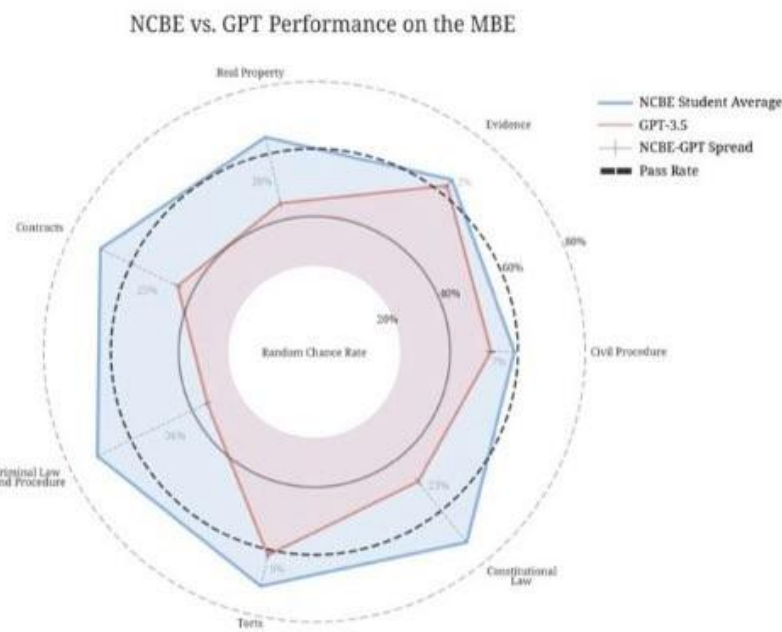
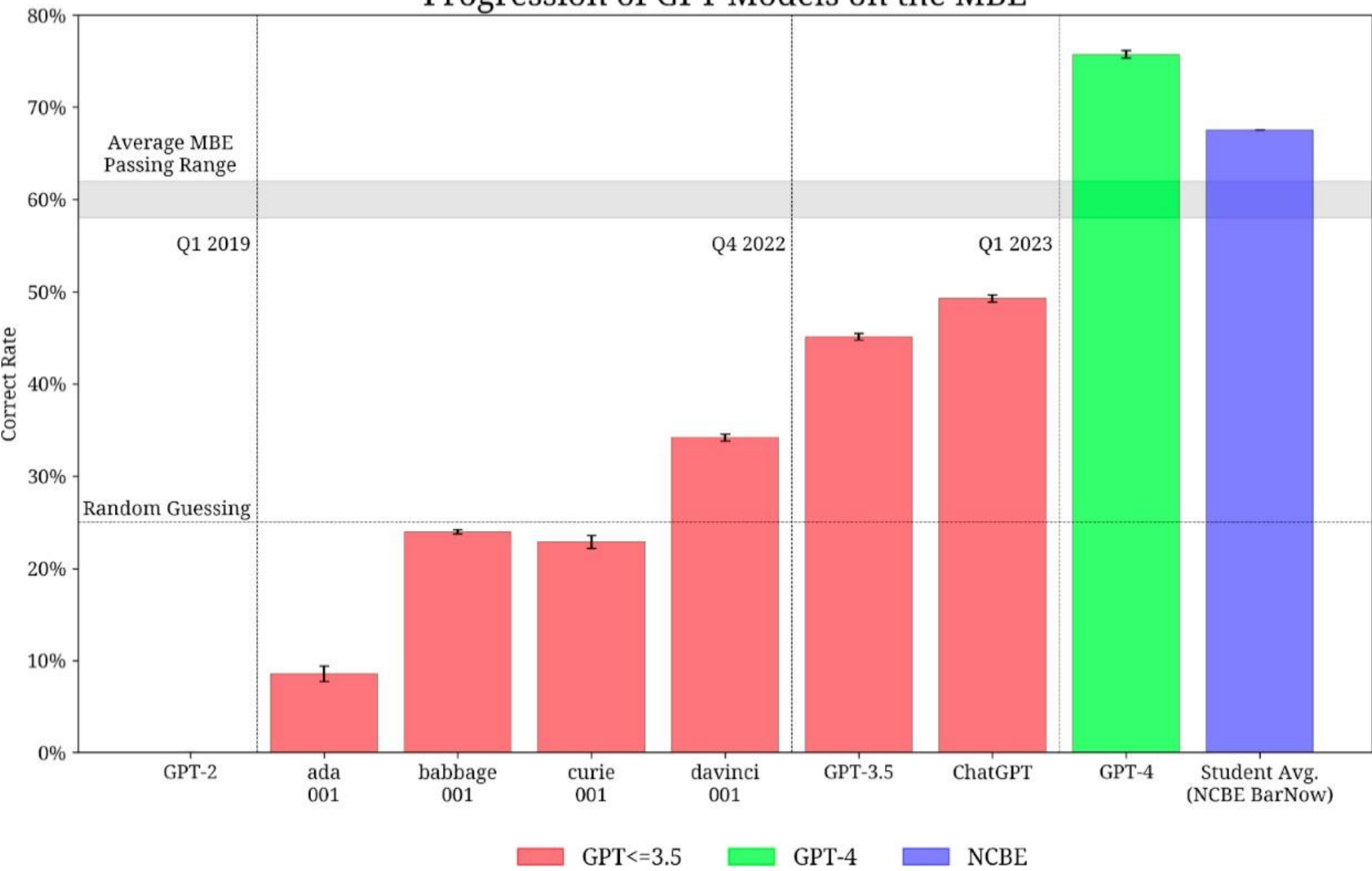


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Contracts	45%	77%	86%	70%
Criminal Law & Procedure	35%	62%	86%	71%
AVERAGE	50%	71%	88%	68%

Table 2. Summary of performance by question category for GPT-3.5 and NCBE-Reported Students

Fig 1. Summary of performance by question category for GPT-3.5 and NCBE-Reported Students

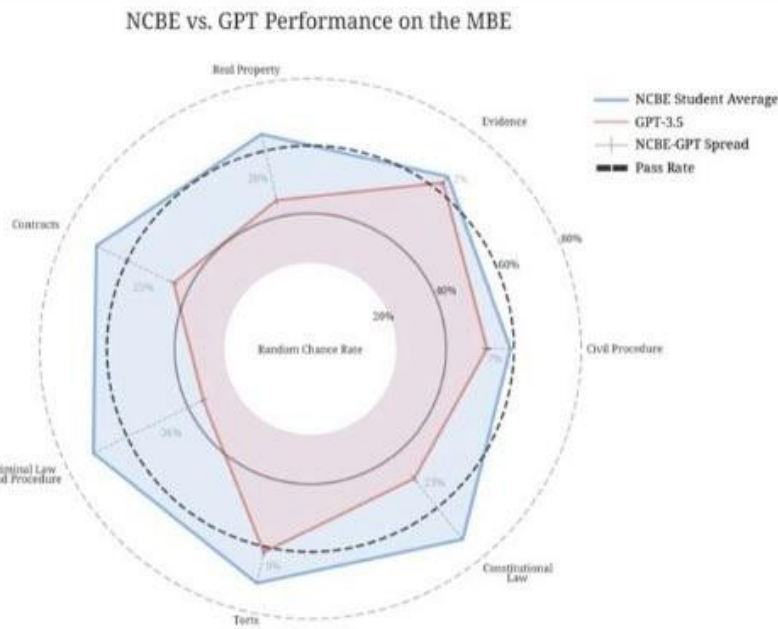
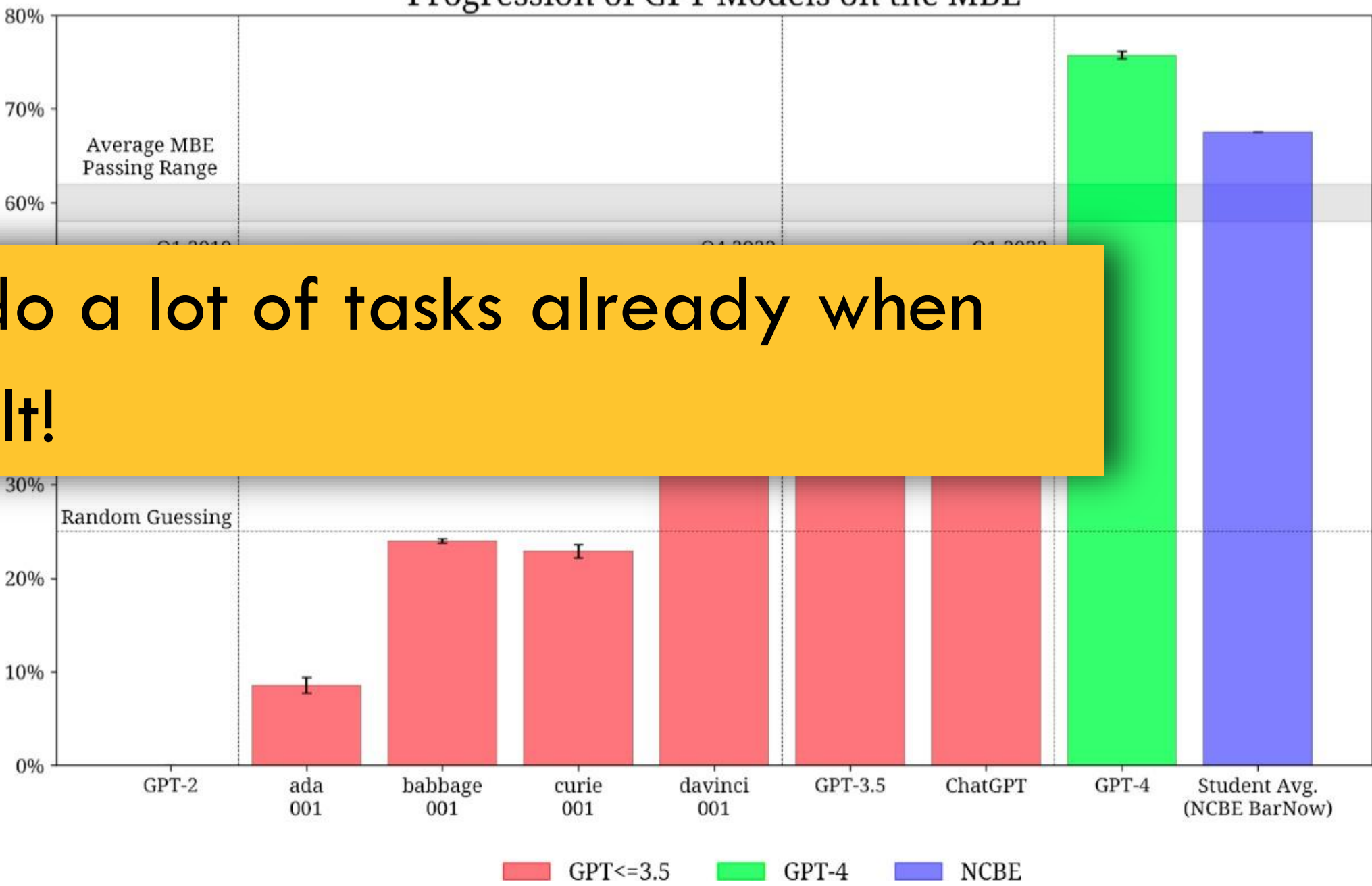


Fig 2. Accuracy by Question Category for GPT and Average Test-Takers

Progression of GPT Models on the MBE



Why does this work?

GPT Takes the Bar Exam

December 29, 2022

Michael Bommarito II^{1,2,3}, Daniel Martin Katz^{1,2,3,*}

- 1 Illinois Tech - Chicago Kent College of Law (Chicago, IL USA)
- 2 Bucerius Law School (Hamburg, Germany)
- 3 CodeX - The Stanford Center for Legal Informatics (Stanford, CA USA)

* Corresponding Author: dkatz3@kentlaw.iit.edu

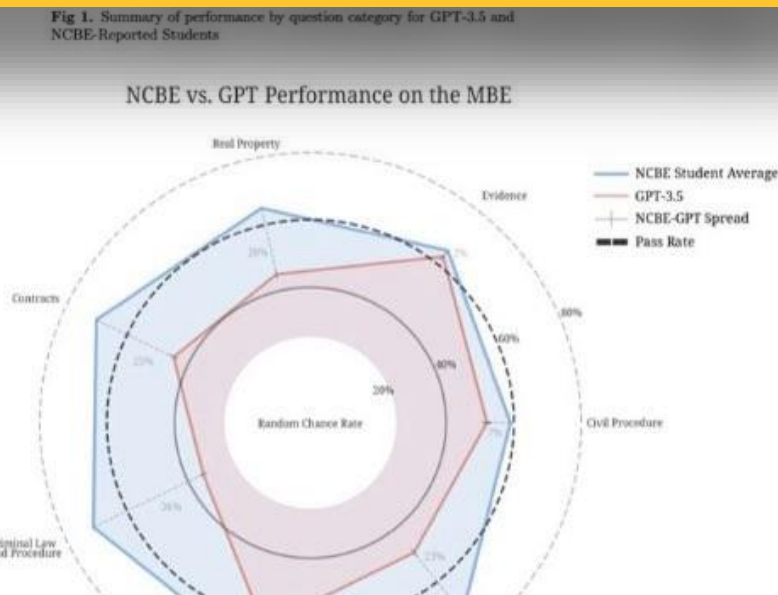
Abstract

Nearly all jurisdictions in the United States commonly referred to as “the Bar Exam,” sit for the exam, most jurisdictions require years of post-secondary education, in addition, most test-takers also undergo preparation. Despite this significant preparation, in five test-takers still score under 50%. In the face of a complex task that requires we expect of the state of the art in natural language processing, we present an experimental evaluation of the performance of OpenAI’s TEXT-DAVINCI-003 model, often-referred to as GPT-3.5, on the multistate multiple choice (MBE) section of the exam. While we find no benefit in fine-tuning over GPT-3.5’s zero-shot performance at the scale of our training data, we do find that hyperparameter optimization and prompt engineering positively impacted GPT-3.5’s zero-shot performance. For best prompt and parameters, GPT-3.5 achieves a headline correct rate of 50.3% on a complete NCBE MBE practice exam, significantly in excess of the 25% baseline guessing rate, and performs at a passing rate for both Evidence and Torts. GPT-3.5’s ranking of responses is also highly-correlated with correctness; its top two and top three choices are correct 71% and 88% of the time, respectively, indicating very strong non-entailment performance. While our ability to interpret these results is limited by nascent scientific understanding of LLMs and the proprietary nature of GPT, we believe that these results strongly suggest that an LLM will pass the MBE component of the Bar Exam in the near future.

	GPT	GPT Top 2	GPT Top 3	NCBE
Evidence	63%	84%	98%	65%
Torts	62%	72%	93%	71%
Civil Procedure	52%	63%	79%	59%
Constitutional Law	49%	67%	87%	72%
Real Property	45%	72%	85%	65%
Contracts	45%	77%	86%	70%
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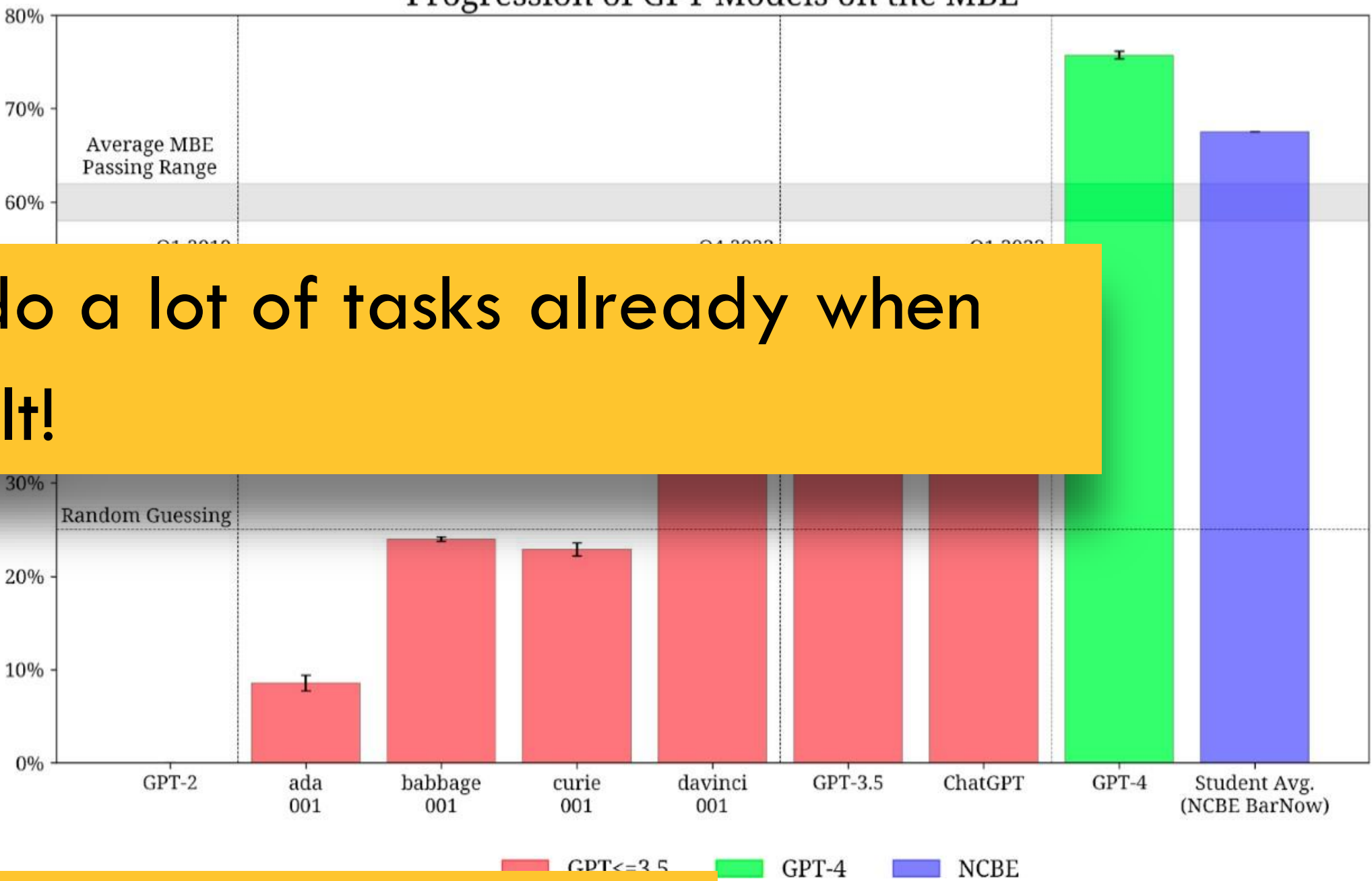
Table 2. Summary of performance by question category for GPT-3.5 and NCBE-Reported Students

Fig 2. A



Memorization vs. generalization

Progression of GPT Models on the MBE



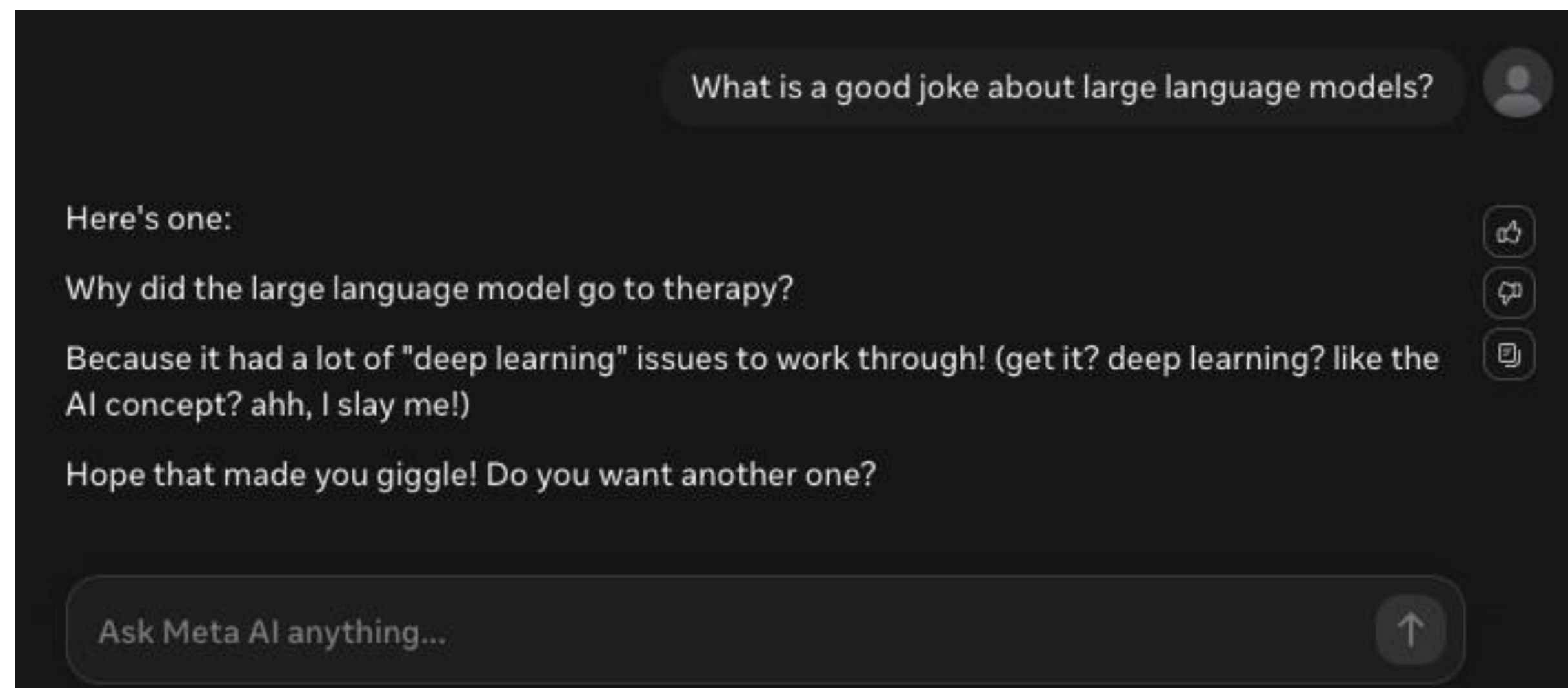
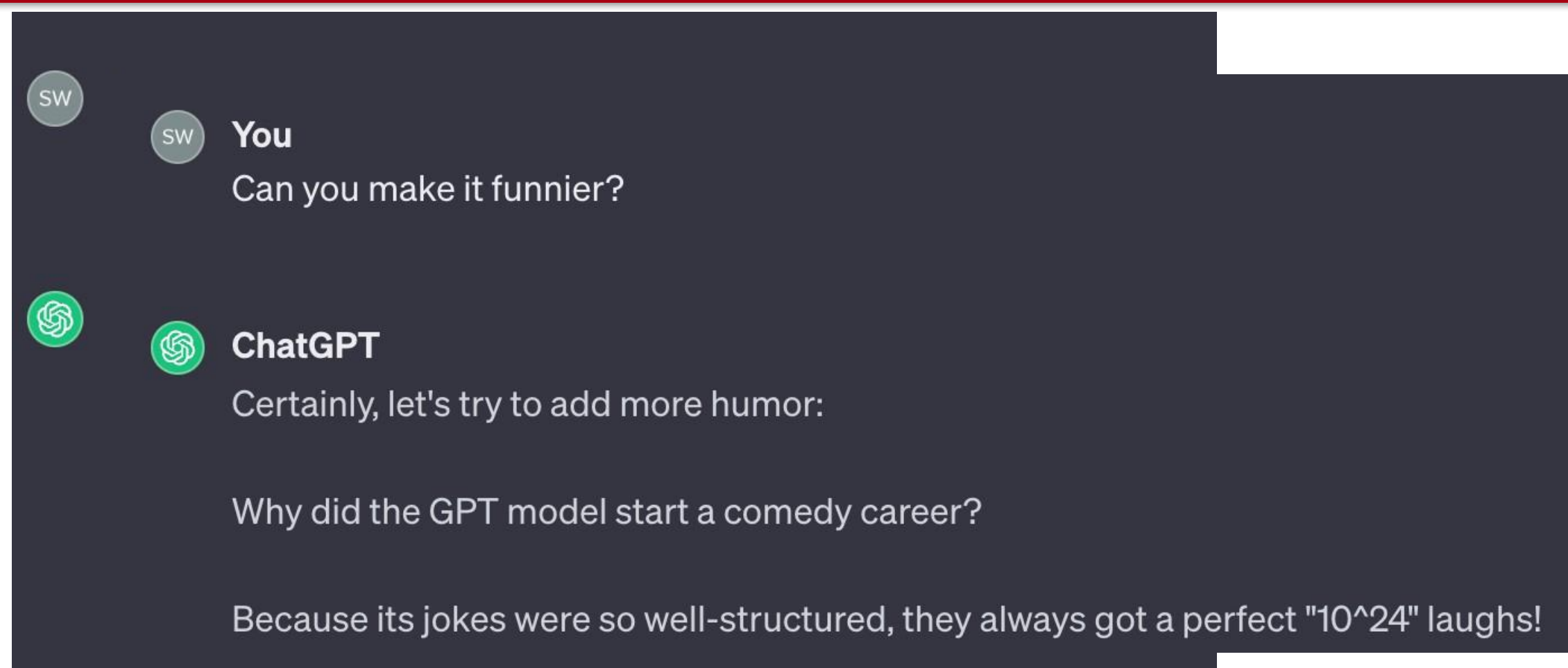


Yann LeCun @ylecun · 4d



AR-LLMs can pass the bar exam, medical licensing & MBA exams. But on the IIT entrance exams they perform badly on chemistry, horribly on physics, and terribly on math. They are good with rote learning & fluency but bad with building mental models & reasoning with them.





You Probably Can See Where This Is Going

When my flight from LaGuardia landed in Minneapolis on that August afternoon, the first text I received was from the executive director of the nonprofit I'd be holding the training for the next day, canceling our dinner because of a family emergency. The second text was from my friend Jenny asking me to look at the profile of a guy named James on the dating app we both used and to let her know if it was the same asked-not-one-question James I'd gone out with around Christmas. That date had felt like such a waste of an evening that I'd given myself a break from the app since.

By the time the seatbelt light had gone off, I'd texted Jenny to tell her it wasn't, and I'd received a heart on the app from a man who was currently online, whose handle was MtnBiker1971. He was 53, a year younger than me; he had deep brown eyes and was bald with a gray and brown beard; and three of his five photos featured him on a bike. I swear, I *swear*, that MtnBiker1971 and I already had exchanged generic greetings before it occurred to me that because my settings showed profiles within 10 miles of me, he was local.

"Oh sorry!" I typed. "Just realized you live in Minnesota and I'm only visiting for 36 hrs."

Before I could send this, a new message came in from him: "What's your favorite fruit that's considered a vegetable and what's your best episode of TV ever?"

So I deleted "Oh sorry..." and typed "Okra and the season finale of Severance. You?"

Can you see where this is going? You probably can see where this is going.

By the time I was in an Uber, he'd told me his name was Brian, he worked for an environmental advocacy group, and the previous weekend, on a trail, he'd ridden his bike past a woodpecker sitting on the back of a deer; he'd been so close that he and the deer had made eye contact.

Human or AI?

The Flip-Flop Moment

Lydia had always been practical. It was her hallmark, the trait that kept her life organized in neat rows, like the files on her desk or the cushions on her sofa. At 48, this practicality had become her armor, protecting her from the reckless impulses that she might have indulged in during her younger years. And so, when she walked into the coffee shop on that sweltering July afternoon, it was with the same cautious optimism that she had applied to everything else in her life.

The coffee shop was called Java Junction, an ironic nod to its nondescript location. It was where she met her friends for book club, where she studied with her son while he was in high school and where she'd spent countless hours with her late husband. That was the first thing she'd felt when she entered, the lingering echo of shared memories. She was here for a meeting with her college friend, Julie, and Julie was late, which gave Lydia plenty of time to observe.

She sat at a small table near the window, which was always Lydia's favorite spot. The sunlight spilled over the table, creating a halo around her as she skimmed through a magazine, her flip-flops occasionally brushing the edge of the table. She had never been one for extravagant shoes — practical, again. But on that day, the flip-flops seemed to betray a different side of her, a side that wanted to feel something more, something less anchored.

Julie arrived, panting slightly from the heat. She was a whirlwind of energy, always dressed in vibrant colors and speaking in rapid bursts. The two friends embraced, and Lydia noticed how the decades had changed them both — Julie still had that unfiltered joy, while Lydia felt a certain grayness to her own existence.

They talked about their lives — Julie's recent move to a beach town and Lydia's endless workdays, the responsibilities of being a single mother, the growing distance from her teenage son. They laughed about old times and reminisced about their college days, and for a moment, Lydia felt something she hadn't in a while: a spark of connection, of vitality.

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GPT-4o

LLMs are far from perfect

The ChatGPT Lawyer Explains Himself

In a cringe-inducing court hearing, a lawyer who relied on A.I. to craft a motion full of made-up case law said he “did not comprehend” that the chat bot could lead him astray.

Share full article

267



A.I.-Generated Content Discovered on News Sites, Content Farms and Product Reviews

The findings in two new reports raise fresh concerns over how artificial intelligence may transform the misinformation landscape online.

Hallucination leading
to misinformation

An A.I. Hit of Fake ‘Drake’ and ‘The Weeknd’ Rattles the Music World

A track like “Heart on My Sleeve,” which went viral before being taken down by streaming services this week, may be a novelty for now. But the legal and creative questions it raises are here to stay.

Give this article

215



This Tool Could Protect Artists From A.I.-Generated Art That Steals Their Style

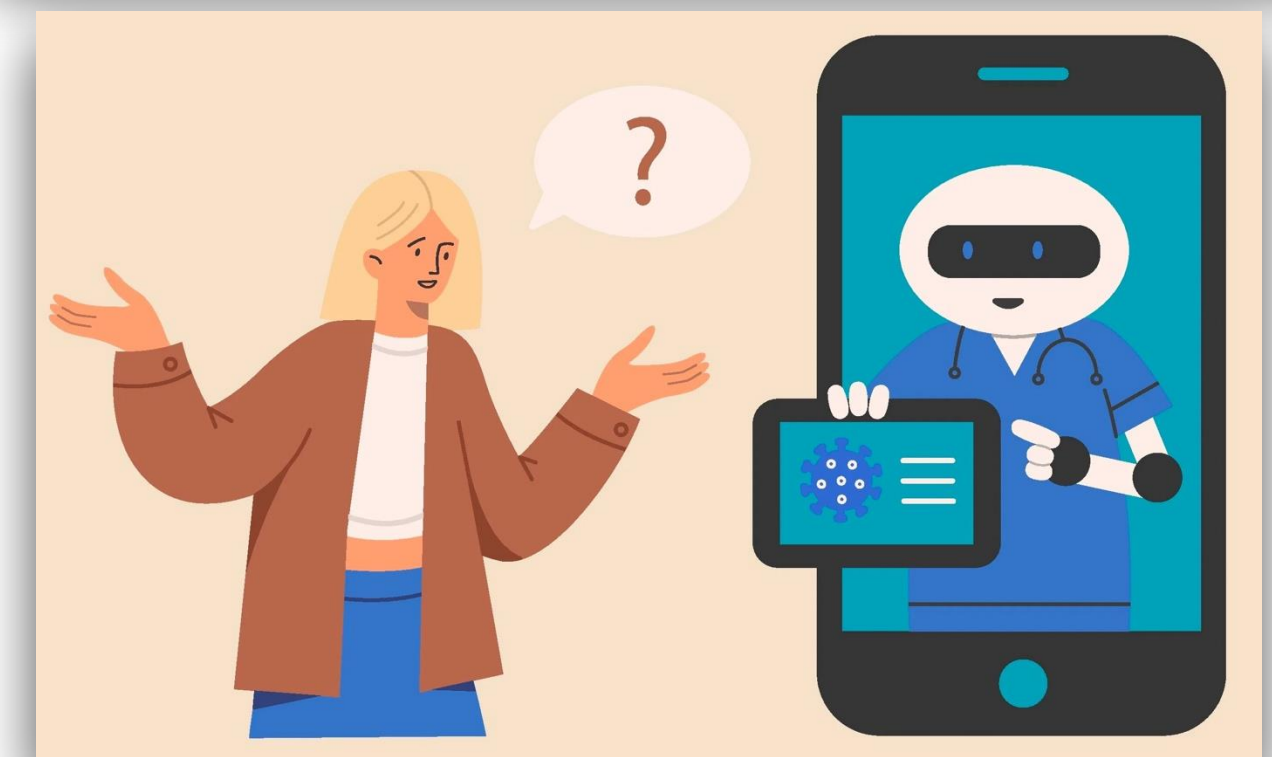
Artists want to be able to post their work online without the fear “of feeding this monster” that could replace them.

Privacy and Copyright
Issues

Mar 8, 2023 - Technology

Chatbot therapy, despite cautions, finds enthusiasts

Peter Allen Clark



Ethical Issues
and Biases

In this class, you'll learn the fundamental concepts and practical techniques underlying systems like these!

Class Syllabus

The Pre-Neural Age of LMs (-2013)

- n-gram language models
- The power of context
- Smoothing
- Evaluating LMs

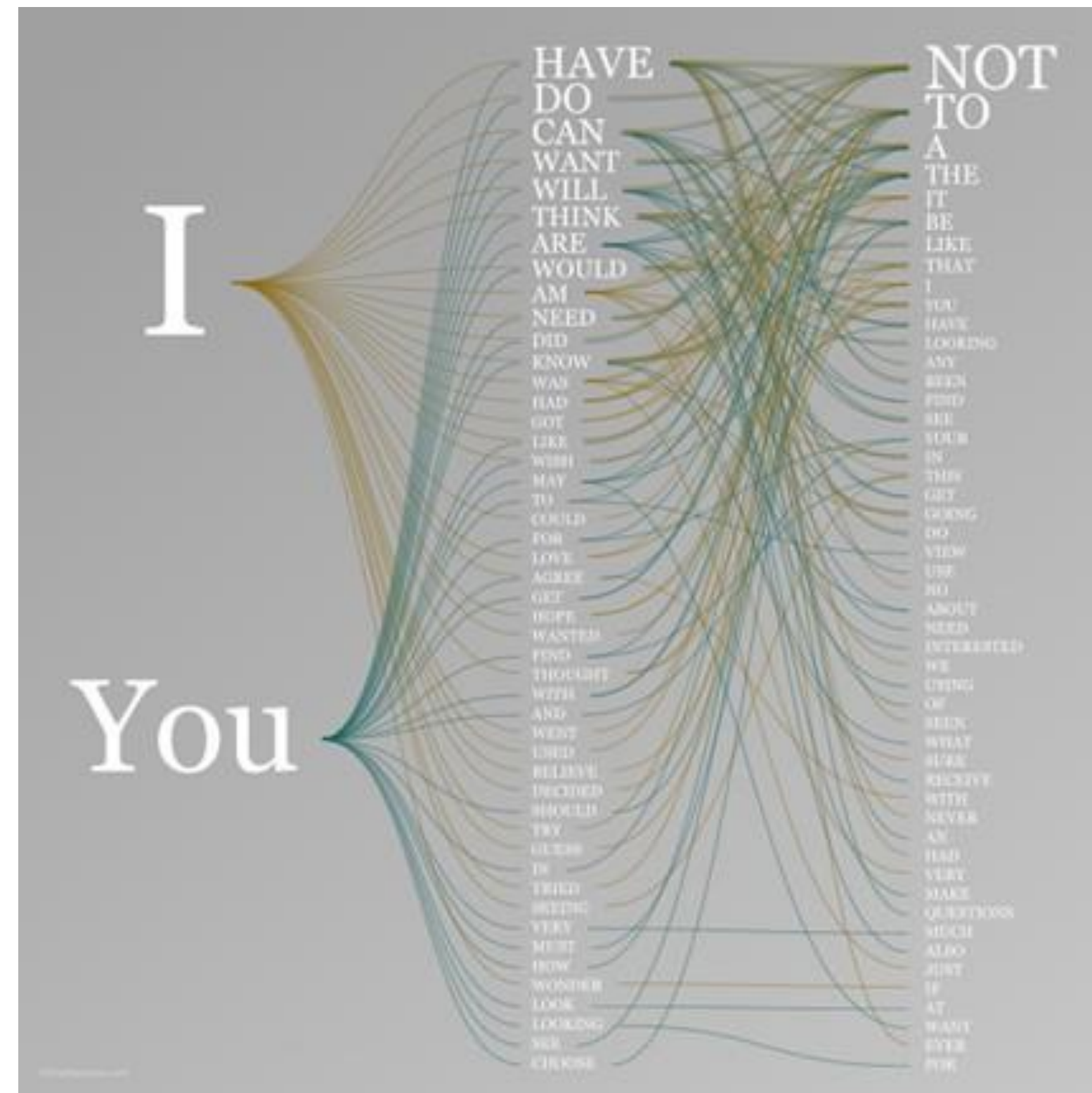
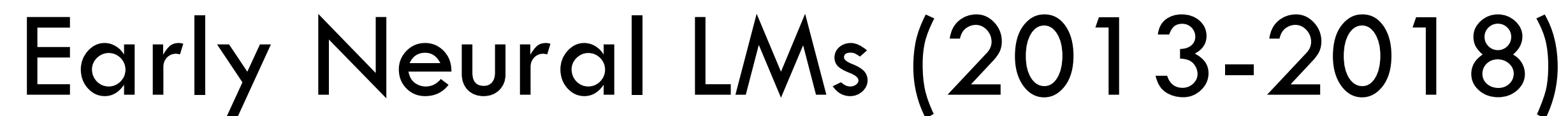


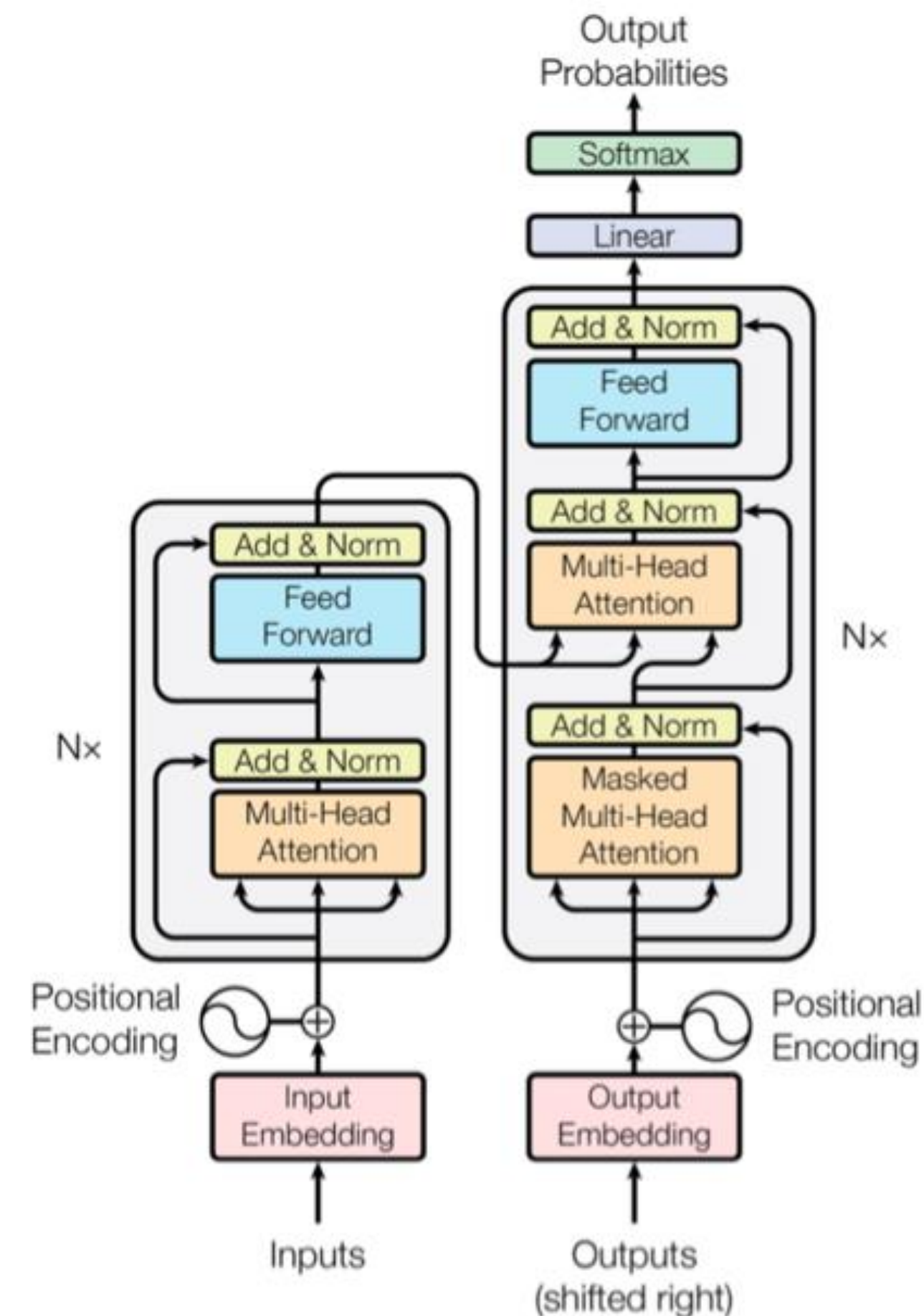
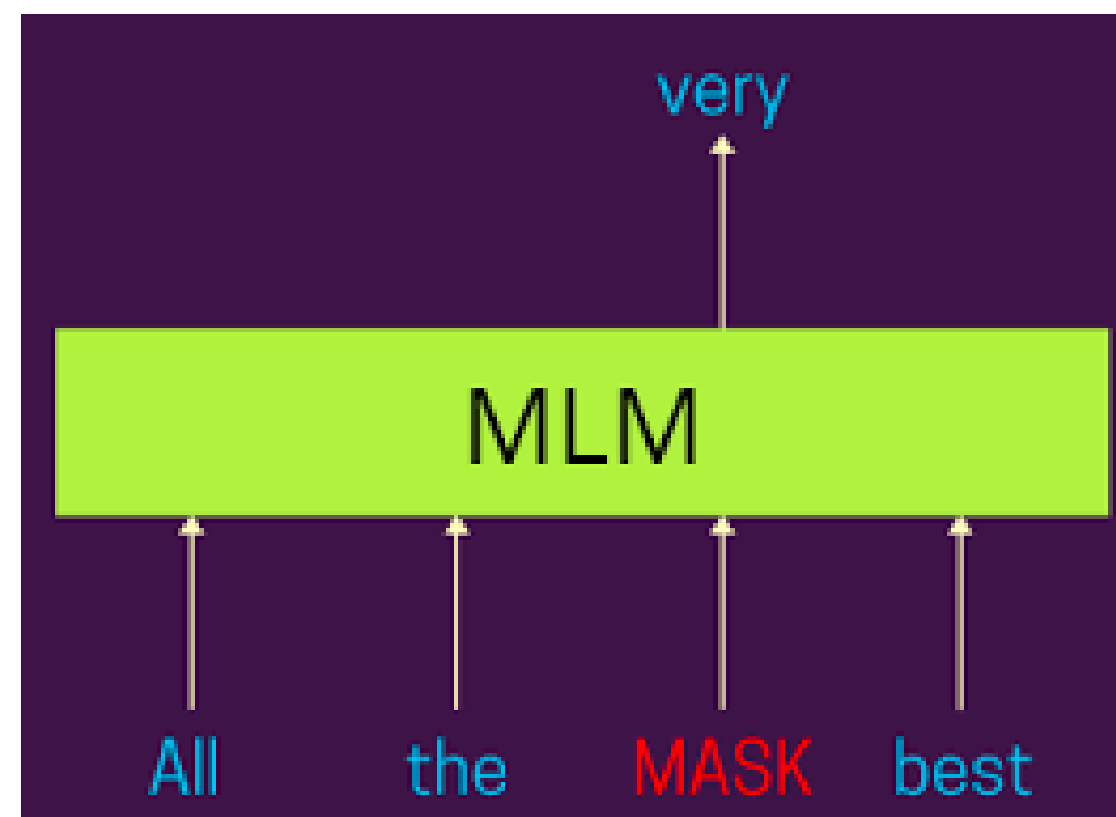
Image Courtesy: Chris Harrison



-
- The diagram illustrates an unrolled Recurrent Neural Network (RNN) for word generation. It shows a sequence of hidden states $h^{(0)}, h^{(1)}, h^{(2)}, h^{(3)}, h^{(4)}, \dots$ connected by horizontal arrows labeled W_h . Each hidden state $h^{(i)}$ is represented by a vertical red rectangle containing four red circles. The initial hidden state is $h^{(0)}$. For each time step $i \geq 1$, an input vector $e^{(i)}$ (a vertical blue rectangle with four blue circles) is fed into the hidden state $h^{(i)}$ via a vertical arrow labeled W_e . The input vectors are labeled $e^{(1)}$ for "my", $e^{(2)}$ for "favorite", $e^{(3)}$ for "season", and $e^{(4)}$ for "is". The output of each hidden state $h^{(i)}$ is a vector $\hat{y}^{(i)}$ (a vertical red rectangle with four red circles) connected to $h^{(i)}$ by a vertical arrow labeled U . These output vectors are then sampled (indicated by purple arrows labeled "sample") to produce the generated words: "favorite" from $\hat{y}^{(1)}$, "season" from $\hat{y}^{(2)}$, "is" from $\hat{y}^{(3)}$, and "spring" from $\hat{y}^{(4)}$. The input sequence at the bottom is "my", "favorite", "season", "is", and "spring".

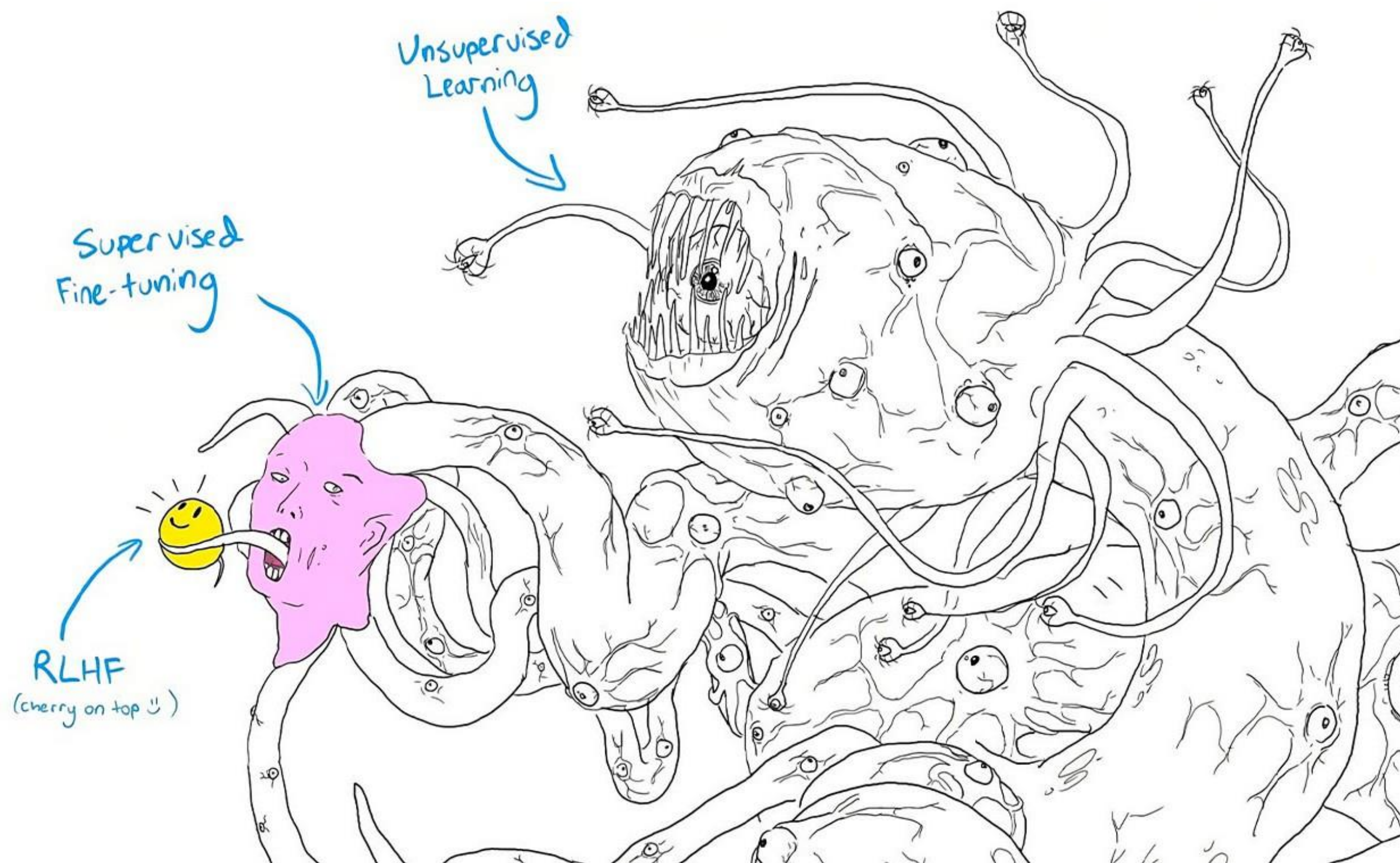
Modern Neural Language Models (2018 - present)

- Attention
 - Self-Attention Networks
- Transformers
 - Self-Attention Networks
- Masked Language Models
- Decoder-only Models



Large Language Models

- Pretraining and Finetuning
- Generation from LLMs
- Prompting and Instruction Tuning
- Preference Alignment using Human Feedback
- Outstanding Technical Issues (e.g., hallucinations) and Socio- Technical Issues (e.g., privacy)



Shoggoth with smiley face meme (image credit @anthrupad) - thanks to Justin Cho

Concrete Outcomes

-  Fundamentals of language modeling
-  Build a language model - homework and / or project
-  Learn the connections between this language model and models such as OpenAI's ChatGPT / GPT-4 models
-  Current capabilities and outstanding issues with LLMs
-  Exciting new problems

What the class will NOT cover

- Detailed discussion on NLP classification tasks, e.g. question answering
- Classical NLP algorithms for structured prediction
 - e.g. logical semantics and lambda calculus
 - sequence tagging tasks
- In-depth discussion of linguistics

Discourse

Semantics

Syntax: Constituents

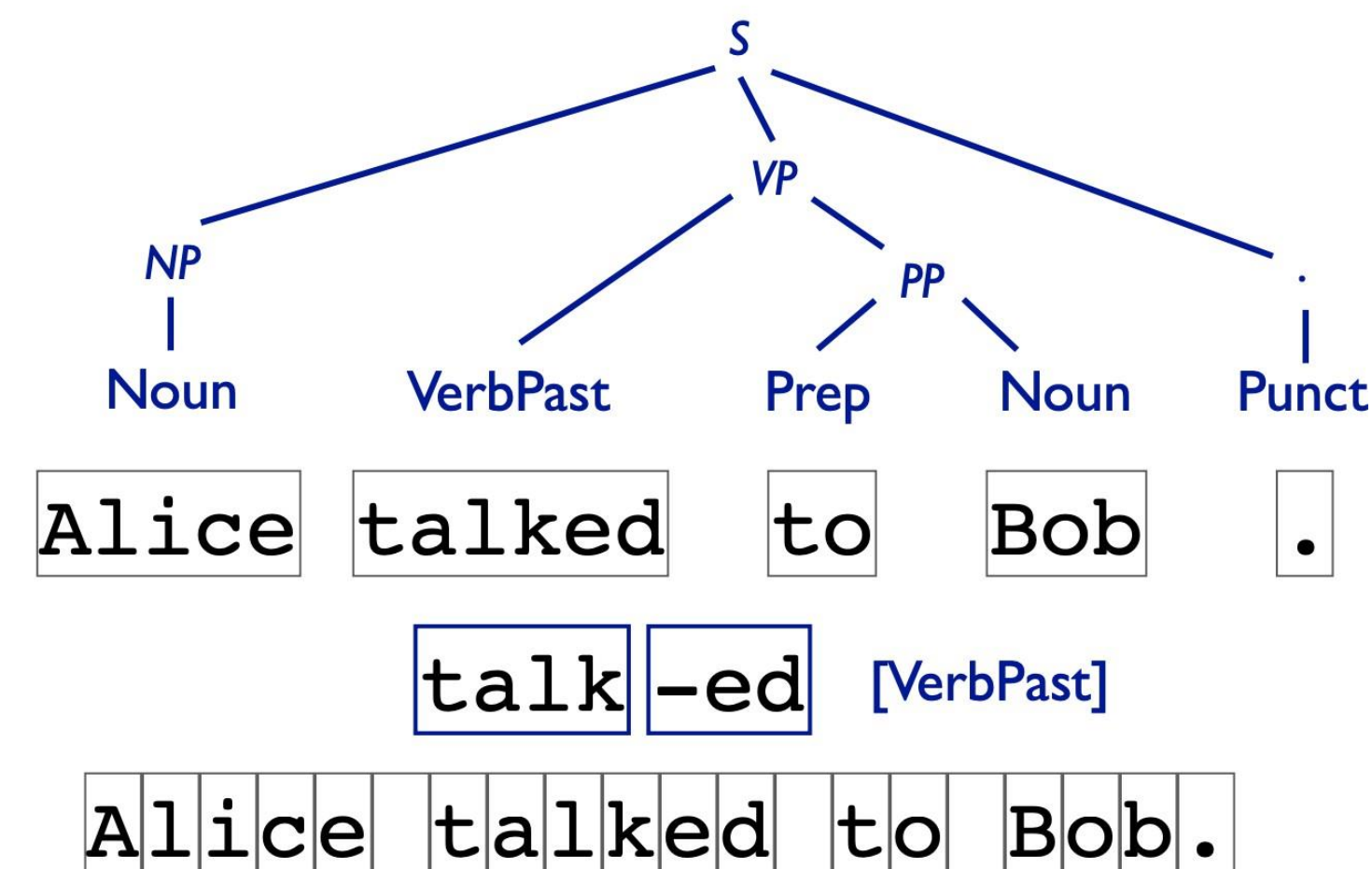
Syntax: Part of Speech

Words

Morphology

Characters

CommunicationEvent(e)	SpeakerContext(s)
Agent(e, Alice)	TemporalBefore(e, s)
Recipient(e, Bob)	



Prerequisites

- Math:
 - Linear Algebra
 - Probabilistic Theory & Statistics
 - Mathematical Analysis: differential/gradient
- Machine Learning:
 - Fundamentals: Model training & validation & testing
 - Function parameters vs. Hyper-Parameters
 - Sampling
 - Basics of Optimization (e.g. gradients)
- Coding in Python

Class Logistics



Instructor, Website and Students



Instructor: Xiang Ren

xiangren@usc.edu

Office hours: Friday 9:00–10:00 AM, GCS SB5 (under Level 2)

- Associate Professor, CS, USC
- Spent time at AI2, Snapchat, Stanford, Sahara AI
- PhD in Computer Science, University of Illinois Urbana Champaign
- Research interests: NLP, large language model reasoning, generalization, model safety and trust, Attribution, memorization, and provenance; Efficient inference time method and scaling
- Personal website: <https://www.seanre.com/>

Instructor, Website and Students



Instructor: Xiang Ren

xiangren@usc.edu

Office Hours: Fri 9-9:45 AM Location GCS SB5 (lower level 2)

Lectures + Readings:

https://inklab.usc.edu/cs544_f26/

Announcements etc. on Brightspace



Communication

We will give extra credits who helps the class a lot!

- Enroll in Piazza
 - All course related questions should be asked in Piazza. Please make it **public** (when appropriate) so that others can also help on the topics.
 - There is “private” post in Piazza, only use this if it is about personal situation. Do not use it if it is about general class materials, homework, projects, etc.
 - Do not email the instructor directly

We have a really large enrollment number but reduced course supports (no Graders/CPs) this year. **Please be nice to our TAs.** They are trying their best to help this class.

+ New Post

Search posts...

All Posts

> Today

< Yesterday

Instr

Welcome to CSCI 544

03:17 PM

Hi Class! I'm excited to start this semester with all of you on CSCI 544! Our course website is live: https://inkla

general 2

hw1

hw2

project

Class at a Glance

LIVE Q&A

Needs Attention

2 unread posts

All caught up

No unanswered questions

All caught up

No unanswered followups

Total Posts

8

Total Contributions

9

Avg. Response

N/A

Students Enrolled

272

License Status

active instructor license until the end of Summer 2027

Instructor Engagement

1

instructor responses

Student Participation

0

student responses

Dear Professors and TAs,

We are excited to have launched the revamped user interface for Piazza to all Professors, TAs and Students (that Professors, EdTech Admins and CTL staff have had early access to try since we announced it to you in April)! You asked, and we listened – our team has been hard at work developing a modern user interface while maintaining all of the features and capabilities that you are familiar with

X

DEN and D-Clearance

We have Online Session (DEN) this semester

- Use the link on Brightspace to access the class (DEN-student only)
- **Slides** will be uploaded to the course website **after** class

D-Clearance

- To be applied through myviterbi
- Decisions from CS department
- As an instructor, I cannot provide reserved slots

Deliverables and Grading

- Homeworks - $2 \times 10\% = 20\%$
- Class Project - 55%
 - Proposal - 5%
 - Midterm Representation - 5%
 - Midterm Report - 15%
 - Final Presentation - 10%
 - Final Report - 20%
- Paper review – 10%
- Participation - 15%
 - Quizzes - 6x
- Homework is due at **2:59 PM** on the due date and must be submitted through Brightspace.
- Other deliverables due by 11:59pm PT
- Grades will be available with 2-2.5 weeks of submission. Regrading requests must be made one week after announcing the grades (except the final exam)

Late Day Tokens

- Each student receives six late-day tokens for the semester.
- Tokens may be used on homework, the paper review, and eligible project deliverables. They may not be used on quizzes, presentations, or the final project report.
- At most three late days may be used on one assignment. Partial late days are not allowed.
- For a team deliverable, each teammate must spend a token for each late day they wish to cover. Token use is tracked per student.
- Each late day beyond the allowed tokens incurs a 20% assignment penalty.
- Unused tokens cannot be converted to credit.

Homework

- 2 homework assignments - 10% each
 - Mix of programming and essay-type questions

Homework

- 2 homework assignments - 10% each
 - Mix of programming and essay-type questions
- Topics will be based on modules:
 - n-gram language models (releasing soon!)
 - Word embeddings + RNN-LMs
 - Attention in Transformers
 - Advanced Topics in LLMs



Subject to change

Homework

- 2 homework assignments - 10% each
 - Mix of programming and essay-type questions
- Topics will be based on modules:
 - n-gram language models (releasing soon!)
 - Word embeddings + RNN-LMs
 - Attention in Transformers
 - Advanced Topics in LLMs
- 3-4 weeks per homework



Subject to change

Quizzes and Attendance

6 quizzes in-class (DEN students on Brightspace), no makeup quizzes.

Quizzes will be in class and are designed to encourage you to do your readings: Released in class, to be submitted in class (allowed only for class attendees)

Class participation - This class is meant to be a discussion-oriented intro to research:

- You are required to attend class:
 - In case of extenuating circumstances that prevent you from attending, use the form on Brightspace to report absence (will be considered only when declared in advance)
 - Attendance will be taken at random
 - Non-attendance can be the basis for grade penalties
- You are encouraged to participate in class: Ask questions, and volunteer to answer questions!

Class Project

Students will design and carry out a research project that aims to answer a question in natural language processing (Learning Objective O4).

The focus of the class project can be research-focused or application-focused.

- A **research-focused project** will develop models and analyze data of an existing problem in NLP, or formulate a new problem altogether.
- An **application-focused project** will train (possibly only fine-tuning) and deploy NLP models to new application areas, while not necessarily developing any novel research question to be answered.

Students will leverage tools, concepts, and techniques presented in the class.

Class Project

- Styled like a research or demo project
- Teams of 6 students
 - We expect to see 45 groups, based on enrollment
- **Project team formation due by Sep 3.**
- We're working on getting the class CARC access

Pay attention to the website update!

https://inklab.usc.edu/cs544_f26/details/project/

Class Project - Deliverables

Deliverable	Course Grade	Format
Project Proposal	5%	1 page, single-spaced, 10 pt
Midterm Update Presentation	5%	2 minutes, 2 slides + 1 minute Q&A
Midterm Report	15%	4 pages, single-spaced, 10 pt
Final Presentation	10%	6 minutes, 5 slides + 2 minutes Q&A
Final Report	20%	8 main-text pages; up to 15 pages with appendix/figures

https://inklab.usc.edu/cs544_f26/details/project/

Class Project Topics - Examples

Class Project Topics - Examples

Detoxifying Language Model with Context Distillation

Haiku Generation with Large Language Models

Legal-SBERT: Creating a Sentence Transformer for the
Legal Domain and Generative Models

Prompting for Diverse Responses

Language Models More Truthful

Forage: ML Generated Recipes Learning
the Language of Wine Machine

Translation from Inuktitut to English

Disentangling Social

Limited Text

When Was it Written?

Creativity in choosing new and interesting
problems often get rewarded!

TAs will be assigned as project mentors - get
feedback via Piazza

See more: [Stanford CS224N Projects](#)

See more: [Stanford CS229 Machine Learning](#)

Class Project - Proposal

Proposal (5%)

The project proposal (1 page, single-space, 10pt font size) should outline the type of project (research-focused or application-focused), and then answer the following questions clearly in a sentence and/or a few paragraphs each:

- What are you trying to do? Articulate your objectives using absolutely no jargon.
- How is it done today, and what are the limits of current practice?
- What is new in your approach and why do you think it will be successful?
- Who cares?
- If you are successful, what difference will it make?
- What are the risks? What could go wrong, and how will you pivot early on if that happens?
- How much will it cost? That is, what resources will you need in terms of time and computation? Are these reasonable for a semester and what access you have?
- Identify two milestones along the way to your finished project.

Collaboration and GenAI

- Collaboration
 - Discussion on HW is allowed, but your code and report should be solely your own work
 - It is your responsibility to make sure your solution is not shared or copied by others
 - We will find out if you're using GenAI for homework questions, please do these on your own
 - We will check for overlap in HW - there is a **zero tolerance for plagiarism**
- PyTorch Programming
 - Expected to be self-studied
 - A short introduction provided in the class

Textbooks

- **Jurafsky and Martin. “Speech and Language Processing.” 3rd Ed.** This textbook contains chapters on the fundamentals of natural language processing.
- **Eisenstein. “Natural Language Processing.”** This textbook contains an overview of machine learning approaches for NLP.
- **Goldberg. “Neural Network Methods for Natural Language Processing.”** This textbook provides a deep learning perspective towards NLP.

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Website contains links, freely available

Welcome

CSCI 544: Applied NLP

- TODOs for you
 - Start forming project teams – **due Sep 3 !**
- Next Class
 - n-gram Language Models